

One Equation, Many Posets

Polynomial maps over fields as resources

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Abstract

A map implements another map when it can be placed between an input processor and an output processor. We tell the story of what follows from the single equation $f = h_{\text{out}} \circ g \circ h_{\text{in}}$. We begin with two classes of affine maps $K \rightarrow K$, then three classes of quadratic maps $\mathbb{F}_3 \rightarrow \mathbb{F}_3$, and then the first branching Hasse diagram for quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$. The same language leads naturally to homogeneous forms, projective zero loci, and projective rational maps.

After these examples we give the general framework for polynomial maps $K^n \rightarrow K^m$, with affine input and output processors. Invertible processors produce equivalence classes; singular processors produce a degeneration order; Cartesian powers produce asymptotic exchange rates. The familiar classification of quadratic forms—rank, radical, isotropy, discriminant, and the characteristic-two trace invariant—then appears as the first classical instance of the framework. We draw the resulting posets over finite fields, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_2$, and $\mathbb{Q}_3$, and include larger homogeneous and cubic computations.

The companion paper solved a fiber-signature exchange theory by partition functions and Gibbs curves. Those formulas can now be evaluated on many of the polynomial classes in this paper. Independently of that formula, composition of any operational exchange rates proves $M_{ij}M_{jk} \leq M_{ik}$, so every closed conversion cycle has return at most one. We explain what cycle geometric means can measure and why this no-arbitrage property does not imply positive semidefiniteness. One pair of nondegenerate binary forms is settled exactly: over $\mathbb{R}$ the inertia monotone gives $C_{\text{aff}}(x^2 - y^2 \rightarrow x^2 + y^2) = 1/2$, while an explicit three-copies-for-two identity gives $C_{\text{aff}}(x^2 + y^2 \rightarrow x^2 - y^2) \geq 2/3$, so that pair is certified one-way with a round trip that loses at least a factor two.

Evaluating the same observer on fibrations of the affine plane over $\mathbb{F}_q$ gives closed forms for both directed rates against a linear map: the forward rate is an endpoint rate reading the largest fiber, and the reverse rate has a bottleneck at the family-independent temperature $\beta^* = \sqrt{2} - 1$ reading the second moment of the Frobenius traces. Flatness of a fiber signature is equivalent to a permutation-polynomial condition. We state plainly that these are translations of classical statistics rather than new arithmetic: the rate is a function of the signature, so information flows into the matrix and not out of it. What is not a translation is the list of statistics the comparison provably cannot read.

The final sections are an outlook. The Euler product is an exact joint partition function of prime occupation modes for real $s > 1$, but the nontrivial zeros lie outside this ordinary Gibbs regime. Weil's Hermitian matrix E and the directed operational matrix M presently have no proved bridge. The paper therefore offers a simple route to core concepts and a concrete research program, not a claimed connection to the Riemann hypothesis.

Guiding message. *A simple implementation rule produces core concepts. It was introduced for complexity, but it also organizes the classification and comparison of polynomial and analytic map families.*

Keywords. polynomial maps; affine equivalence; degeneration poset; quadratic forms; finite fields; exchange rate; no arbitrage; cycle mean; partition function; local zeta function.

1 Introduction

This paper begins with a very small piece of language. Let f and g be maps. We say that g implements f when

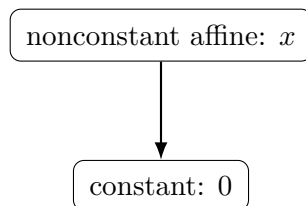
$$f = h_{\text{out}} \circ g \circ h_{\text{in}} \tag{1}$$

for allowed input and output processors. Thus one call to g , together with preprocessing and postprocessing, realizes f . This was the starting point of the finite-map complexity theory in the companion paper [1]. Here we keep the sentence and change the objects: the maps are polynomials over a field, and the main processors are affine.

We will not begin with the most general definition. The first examples are small enough to see completely.

First example: affine maps of a line

Consider degree-at-most-one maps $K \rightarrow K$, with affine input and output processors. Every nonconstant affine map is equivalent to x , and every constant map is equivalent to 0. The map x implements 0 by a constant output processor. A constant cannot implement x . We have already obtained a two-node Hasse diagram:



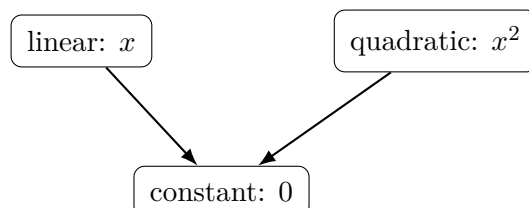
The two core concepts—equivalence class and degeneration order—came from the same implementation rule.

Second example: quadratic maps $\mathbb{F}_3 \rightarrow \mathbb{F}_3$

Every degree-at-most-two polynomial over $\mathbb{F}_3$ is uniquely a function of the form $ax^2 + bx + c$. Invertible affine changes of input and output give three classes:

$$0, \quad x, \quad x^2.$$

Their class sizes are 3, 6, 18, respectively, accounting for all $3^3 = 27$ quadratic polynomial functions of one variable. Both nonconstant classes implement a constant. Neither implements the other: a nonconstant affine substitution preserves the degree of x^2 , while a singular substitution is constant. The poset branches:



This is the first warning that “more complicated” is not automatically a linear ranking. The linear and quadratic classes lose different structure.

Third example: quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$

Now take

$$Q(x, y) = ax^2 + bxy + cy^2 + dx + ey + f.$$

There are $3^6 = 729$ coefficient vectors, but only six affine-equivalence classes:

$$0, \quad x, \quad x^2, \quad x^2 + y, \quad xy, \quad x^2 + y^2.$$

The last two are the split and anisotropic nonsingular quadratic forms. The split form has an isotropic line and can lose its quadratic part along that line, producing x . The anisotropic form has no such line, so it reaches x^2 but not x . Figure 1 later displays the full six-node Hasse diagram with all class sizes.

This example already touches classical quadratic-form theory. The framework has not replaced the classical invariants; it has explained why they appear. Rank, the radical, and isotropy are precisely the information that survives or disappears under singular affine processors.

Why go projective?

Translations mix homogeneous degrees. If we want to study the leading part of a polynomial, or what its fibers do at infinity, the natural next step is to keep only homogeneous maps and use linear processors. A homogeneous scalar polynomial $Q \in \text{Sym}^d(V^*)$ scales by $Q(\lambda v) = \lambda^d Q(v)$, so its zero set descends to projective space $\mathbf{P}(V)$.

There is a small but important distinction. A scalar form on V does *not* give an interesting map $\mathbf{P}(V) \rightarrow \mathbf{P}(K)$, because $\mathbf{P}(K) = \mathbf{P}^0$ is one point. It is naturally a section of $\mathcal{O}(d)$, or equivalently a projective hypersurface $Q = 0$. A tuple

$$F = (F_0, \dots, F_r), \quad F_i \in \text{Sym}^d(V^*),$$

does give a rational map

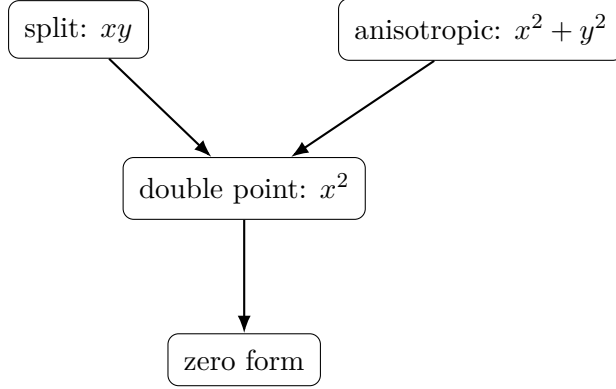
$$\mathbf{P}(V) \dashrightarrow \mathbf{P}^r, \quad [v] \mapsto [F_0(v) : \dots : F_r(v)].$$

It is a total projective morphism exactly when the forms have no common projective zero. Thus scalar forms and tuples belong in the same tensor framework, but their geometric readings are different.

Over $\mathbb{F}_3$, binary quadratic forms have four classes:

$$0, \quad x^2, \quad xy, \quad x^2 + y^2.$$

Their orbit sizes are respectively 1, 8, 12, 6, summing to the $3^3 = 27$ formal binary quadratics. The last two are different degree-two divisors on $\mathbf{P}^1$: split over $\mathbb{F}_3$, or conjugate over $\mathbb{F}_9$. Their degeneration diagram is



Pairs of binary quadratics give rational self-maps $\mathbf{P}^1 \dashrightarrow \mathbf{P}^1$. Their seven classes and eight Hasse covers appear in Figure 6; the base-point-free nodes are genuine quadratic morphisms. Ternary quadratic forms similarly give projective conics in $\mathbf{P}^2$, with five classes. These examples will be drawn in Section 6.

We now return to the general story. Depending on the processor class, the same implementation equation appears as a reduction in complexity theory, a simulation in information theory, a restriction of tensors, or a left–right action in singularity theory. The point is to keep the equation fixed and vary the processors deliberately.

The next continuation is where the companion paper [1] enters. For finite maps with a fiber-preserving processor convention, the asymptotic exchange rate is computed exactly from the partition functions

$$Z_f(\beta) = \sum_y |f^{-1}(y)|^\beta.$$

Many polynomial classes have the same fiber signature, while others do not, so this produces a second numerical geometry on the class poset. It is a useful observer, but it must not be confused with the block-affine exchange rate. The two processor classes are different.

The paper has three levels of certainty.

1. The processor preorder, the quadratic normal forms, the finite-field class counts, and the no-arbitrage theorem are proved or reduced to standard quadratic-form calculations.
2. The larger homogeneous and cubic diagrams, and the displayed exchange matrices, are reproducible finite computations.
3. The possible relation among exchange matrices, local zeta profiles, Weil positivity, and the Riemann hypothesis is exploratory. We state precisely what is known and what is not connected.

This separation is part of the contribution. It prevents a suggestive common vocabulary—resources, partition functions, spectra, exchange matrices—from being mistaken for a theorem that has not yet been found.

2 Processors, equivalence, and order

Fix a field K , dimensions n, m , and a degree bound d . Write

$$\text{Pol}_d(K^n, K^m)$$

for polynomial maps of total degree at most d . An affine input processor and output processor have the form

$$a(x) = Ax + t, \quad b(y) = By + u,$$

with $A \in \text{Mat}_{n \times n}(K)$ and $B \in \text{Mat}_{m \times m}(K)$.

Definition 2.1 (Affine implementation). For $f, g \in \text{Pol}_d(K^n, K^m)$, say that g implements f , and write $f \preceq_{\text{aff}} g$, when

$$f = b \circ g \circ a \tag{2}$$

for affine a and b . The matrices A and B may be singular, including zero.

The convention is that the resource is on the right of $\preceq_{\text{aff}}$: an arrow in a Hasse diagram points from g down to a map f that it implements.

Proposition 2.2. *Affine implementation is a preorder.*

Proof. Identity processors prove reflexivity. If $f = b_1 \circ g \circ a_1$ and $g = b_2 \circ h \circ a_2$, then

$$f = (b_1 \circ b_2) \circ h \circ (a_2 \circ a_1),$$

and affine maps are closed under composition. □

Invertible processors define the affine left–right equivalence relation

$$f \sim_{\text{aff}} g \iff f = b \circ g \circ a, \quad a \in \text{Aff}_n(K), \quad b \in \text{Aff}_m(K) \text{ invertible.} \tag{3}$$

This is the finite-dimensional affine version of the left–right equivalence used for map germs and singularities; see, for example, [4]. The set $\text{Pol}_d(K^n, K^m)$ is a representation space for the invertible affine group, and its equivalence classes are group orbits.

Singular processors induce a directed relation on those orbits. In every quadratic diagram below this relation is antisymmetric. In general it need not be: if two different invertible orbits are mutually reachable using singular processors, one must contract the strongly connected component. The quotient by mutual reachability is always a poset.

Remark 2.3 (Exact reachability versus orbit closure). Our arrows solve the exact equation (2) with possibly singular processors. This is related to, but not automatically the same as, containment of algebraic-group orbit closures. Orbit-closure theory allows limits of invertible transformations and is governed by results such as [5]; an exact singular factorization is a more specific certificate. The two orders should not be identified without proof.

Remark 2.4 (Bounded-degree nonlinear processors). If input processors have degree at most two, a one-step relation is generally not transitive because two quadratic substitutions may compose to degree four. The correct object is then the transitive closure of allowed steps, followed by quotienting mutual reachability. This distinction explains the large collapse in the cubic $\mathbb{F}_3$ example of Section 6.

The implementation order comes with immediate monotones: polynomial degree, the dimension of the affine span of the image, coordinate rank for homogeneous tensors, and appropriate ranks of derivatives or polar forms cannot increase under the corresponding processors. These monotones help prove that proposed arrows are Hasse covers and that omitted arrows are impossible.

3 Cartesian powers and the operational rate

For $f : K^n \rightarrow K^m$, let

$$f^{\times r} : K^{nr} \longrightarrow K^{mr}, \quad f^{\times r}(x_1, \dots, x_r) = (f(x_1), \dots, f(x_r)).$$

The block processor is allowed to mix the r copies. Define

$$k_{g \rightarrow f}(r) = \max\{k : f^{\times k} \preceq_{\text{aff}} g^{\times r}\}.$$

Block-diagonal composition gives $k(r+s) \geq k(r) + k(s)$, so Fekete's lemma gives the limit

$$C_{\text{aff}}(g \rightarrow f) = \lim_{r \rightarrow \infty} \frac{k_{g \rightarrow f}(r)}{r} = \sup_{r \geq 1} \frac{k_{g \rightarrow f}(r)}{r}, \quad (4)$$

with infinite values permitted for degenerate resources.

The same construction works for any processor family that is closed under composition and block sum. The following consequence is the fundamental “no-arbitrage” statement.

Theorem 3.1 (Composition and no arbitrage). *Let $C(i \rightarrow j)$ be operational asymptotic rates in such a resource theory. Then*

$$C(i \rightarrow j)C(j \rightarrow k) \leq C(i \rightarrow k). \quad (5)$$

For normalized nondegenerate resources with $C(i \rightarrow i) = 1$, every directed cycle $\mathcal{C} = (i_0, i_1, \dots, i_\ell = i_0)$ satisfies

$$R(\mathcal{C}) := \prod_{r=0}^{\ell-1} C(i_r \rightarrow i_{r+1}) \leq 1. \quad (6)$$

Proof. Choose block conversions with rates arbitrarily close to $C(i \rightarrow j)$ and $C(j \rightarrow k)$, clear the block denominators, and compose them. This realizes any rate strictly below their product from i to k , proving (5). Iterating around a cycle gives $R(\mathcal{C}) \leq C(i_0 \rightarrow i_0) = 1$. $\square$

For a finite collection put $M_{ij} = C(i \rightarrow j)$. When all entries are positive, the logarithmic costs

$$a_{ij} = -\log M_{ij} \quad (7)$$

satisfy the directed triangle inequality

$$a_{ik} \leq a_{ij} + a_{jk}$$

and every directed cycle has nonnegative total cost. The costs need not be nonnegative edge by edge because one resource may produce more than one copy of another; it is the cycle sum that cannot be negative.

3.1 Why no arbitrage does not imply positive semidefiniteness

The matrix M is directed and normally nonsymmetric, so positive semidefiniteness is not even an intrinsic property of it. More strongly, entrywise positivity, unit diagonal, and all cycle inequalities do not imply positive semidefiniteness even after symmetry is imposed. For example,

$$S = \begin{pmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & 0.1 \\ 0.9 & 0.1 & 1 \end{pmatrix}.$$

Every cycle product is at most one, but

$$\det S = 1 + 2(0.9)(0.9)(0.1) - 0.9^2 - 0.9^2 - 0.1^2 = -0.468 < 0.$$

Thus the no-arbitrage cone and the positive-semidefinite cone are genuinely different geometries.

4 Quadratic maps $K^2 \rightarrow K$

Consider

$$Q(x, y) = ax^2 + bxy + cy^2 + dx + ey + f. \quad (8)$$

For $\text{char } K \neq 2$, write

$$Q(z) = z^\top S z + \ell^\top z + c$$

with S symmetric. Translation by t replaces the linear part by $\ell + 2St$. Hence the component of ℓ in the image of S can be removed. What remains is a linear functional on the radical of S . This gives four preliminary types:

$$\text{constant}, \quad x, \quad x^2, \quad x^2 + y,$$

followed by the similarity classes of nonsingular binary quadratic forms.

The form $x^2 + y$ is called parabolic here. Its linear term lives along the radical of its quadratic part, and no translation removes it. A nonsingular binary form has no radical, so every linear term can be removed.

Under a singular input matrix, a nondegenerate form always restricts to a rank-one form x^2 . It restricts to a nonconstant linear map exactly when it has an isotropic direction: choosing that direction kills the quadratic term while a translation in a transverse direction creates the linear term. This one observation determines the split/anisotropic difference in every diagram below.

Theorem 4.1 (Odd finite fields). *For every odd prime power q , affine equivalence on degree-at-most-two maps $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ has six classes:*

$$1, \quad x, \quad x^2, \quad x^2 + y, \quad xy, \quad x^2 - \nu y^2,$$

where ν is a nonsquare. Their numbers of coefficient vectors are

type	class size
constant	q
linear	$q(q^2 - 1)$
rank-one quadratic	$q^2(q^2 - 1)$
parabolic	$q^2(q - 1)(q^2 - 1)$
split quadratic	$q^4(q^2 - 1)/2$
anisotropic quadratic	$q^4(q - 1)^2/2$

The Hasse covers are

$$\begin{aligned} [x^2 + y] &\longrightarrow [x], [x^2], \\ [xy] &\longrightarrow [x], [x^2], \\ [x^2 - \nu y^2] &\longrightarrow [x^2], \\ [x], [x^2] &\longrightarrow [1]. \end{aligned}$$

Proof sketch. The preceding translation argument gives the four singular types. The usual Witt classification of nonsingular binary forms over $\mathbb{F}_q$ gives one split and one anisotropic similarity class; see [6, 7]. The arrows follow from rank-one restrictions and isotropy. Orbit-stabilizer counts give the displayed sizes, which sum to q^6 . $\square$

Figure 1 is the case $q = 3$. Its six class sizes are 3, 24, 72, 144, 324, 162, summing to $3^6 = 729$.

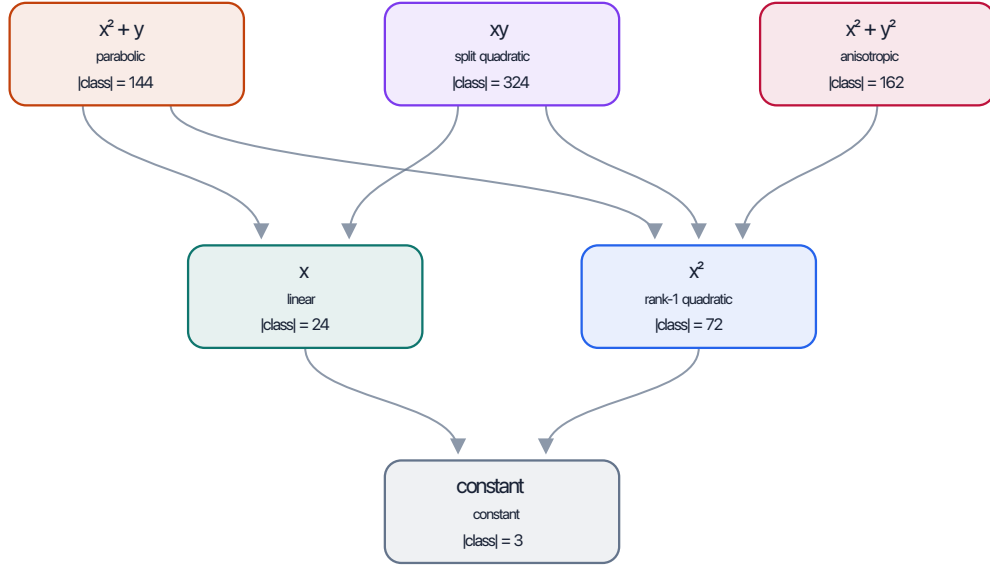


Figure 1: The affine-processor Hasse diagram for quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$: six classes and seven covers. Singular affine input and output processors are allowed. Arrows point from a resource map to an immediately implementable degeneration; class sizes count coefficient vectors.

4.1 Characteristic two

In characteristic two the polar bilinear form no longer determines the quadratic form, and square terms behave additively. Over $\mathbb{F}_2$, polynomial functions satisfy $x^2 = x$ and $y^2 = y$, leaving only three classes: constant, balanced linear, and weight-one-or-three, represented by $1, x, xy$.

For even $q \geq 4$, there are seven classes. In addition to the constant, linear, pure-square, parabolic, split, and anisotropic classes, there is an aligned-linear class $x^2 + x$. With $\delta \in \mathbb{F}_q$ satisfying $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(\delta) = 1$, the nonsingular representatives are xy and $x^2 + xy + \delta y^2$. The class sizes are

type	class size
constant	q
linear	$q(q^2 - 1)$
pure square	$q(q^2 - 1)$
square plus aligned linear	$q(q - 1)(q^2 - 1)$
parabolic	$q^2(q - 1)(q^2 - 1)$
split quadratic	$q^4(q^2 - 1)/2$
anisotropic quadratic	$q^4(q - 1)^2/2$.

Again the sum is q^6 . The trace/Arf invariant replaces the discriminant test. The aligned-linear node is incomparable with the pure-square and ordinary linear nodes, although all three degenerate to a constant.

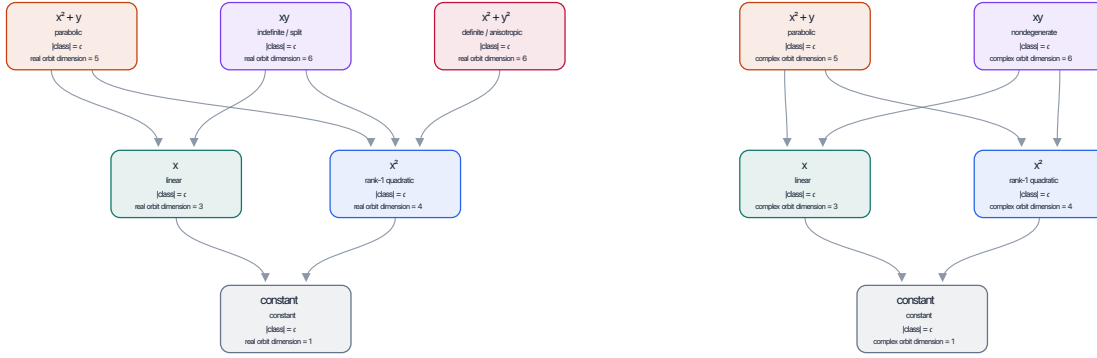


Figure 2: The same affine implementation convention over $\mathbb{R}$ and $\mathbb{C}$. The real diagram has six classes and seven covers; the complex diagram has five classes and six covers. Singular affine input and output processors are allowed. The field changes the nonsingular quadratic-form classes, not the definition of the order.

5 Changing the field

The processor definition does not change with the field; only the quadratic similarity classes and isotropic directions change.

5.1 The real and complex numbers

Over $\mathbb{C}$, every nonsingular binary quadratic form is similar to xy , so the diagram has five nodes. Over $\mathbb{R}$, output rescaling identifies positive and negative definite forms, leaving two nonsingular classes: the indefinite form xy and the definite form $x^2 + y^2$. The real diagram therefore has six nodes. The definite class has no arrow to x , exactly because it has no real isotropic direction.

All nonempty real or complex orbits have continuum cardinality. Orbit dimension is the useful label: constant, linear, rank-one, parabolic, and open nonsingular strata have dimensions 1, 3, 4, 5, 6, respectively, over the base field. These are real dimensions for $\mathbb{R}$ and complex dimensions for $\mathbb{C}$.

5.2 Local fields

For a non-Archimedean local field, the square-class group creates several nonsingular binary forms. Over $\mathbb{Q}_3$, the representatives $d \in \{1, -1, 3, -3\}$ in

$$x^2 + dy^2$$

give one split class $d = -1$ and three pairwise-incomparable anisotropic classes. Together with the four singular types, this gives eight nodes.

Over $\mathbb{Q}_2$, the square classes

$$\{1, -1, 2, -2, 5, -5, 10, -10\}$$

give one split and seven anisotropic classes, hence twelve nodes. Every anisotropic node reaches x^2 but not x , and the seven are pairwise incomparable in the one-shot affine poset. Figure 4 displays the result.

The orbit dimension labels are again 1, 3, 4, 5, 6 in the six-dimensional coefficient space. Cardinality cannot distinguish the orbits because every class has continuum size.

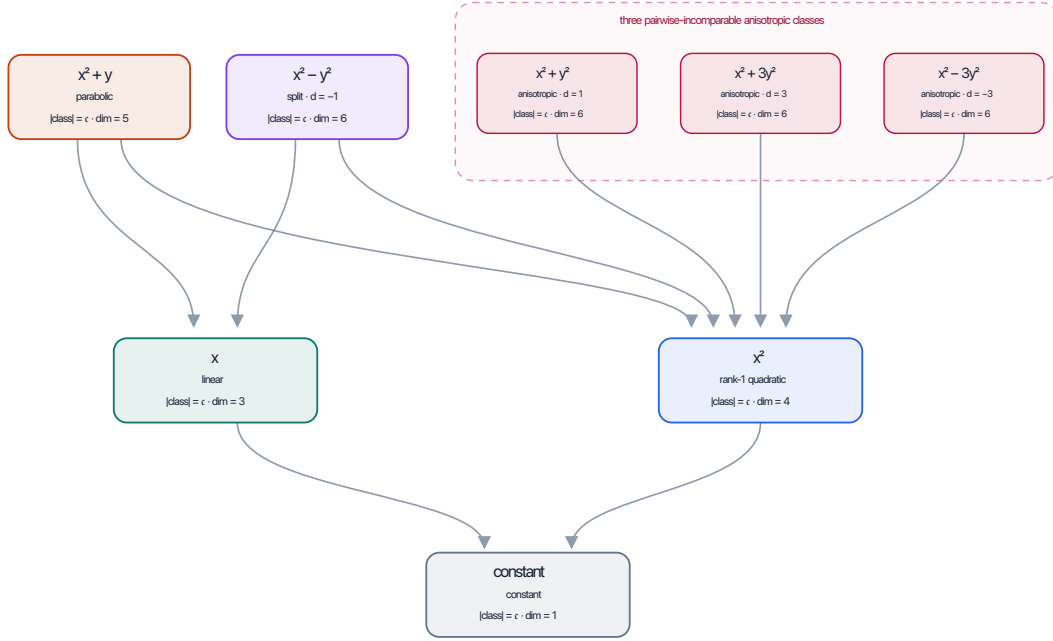


Figure 3: Quadratic maps $\mathbb{Q}_3^2 \rightarrow \mathbb{Q}_3$ with singular affine processors: eight classes and nine covers. The three anisotropic square classes are incomparable and have no arrow to the linear node.

6 Beyond scalar quadratic maps

The same construction applies to homogeneous tensors

$$F \in \text{Hom}(\text{Sym}^d V, W), \quad G = B \circ F \circ A.$$

Here translations are removed to preserve homogeneity. Invertible $A \in \text{GL}(V)$, $B \in \text{GL}(W)$ define equivalence; arbitrary linear maps define degeneration.

Exhaustive calculations over $\mathbb{F}_3$ give the following atlas.

tensor family	total tensors	classes	Hasse covers
ternary quadratic forms	3^6	5	5
quadratic $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	3^6	7	8
quadratic $\mathbb{F}_3^3 \rightarrow \mathbb{F}_3^3$	3^{18}	50	210
ternary cubic forms	3^{10}	26	66
cubic $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	3^8	19	32

The quadratic 3-input, 3-output calculation quotients output changes first, replacing coordinate tuples by subspaces of the six-dimensional space of ternary quadrics. Only 45,256 subspaces of dimension at most three need be enumerated instead of all 3^{18} tensors.

The first row of the table is the projective-conic example promised in the introduction. A ternary quadratic form is a section on $\mathbf{P}^2$; its zero locus is the whole plane for the zero form, a double line, two lines, an anisotropic rank-two conic, or a nonsingular conic. Singular linear input processors contract these projective configurations exactly as shown in Figure 5.

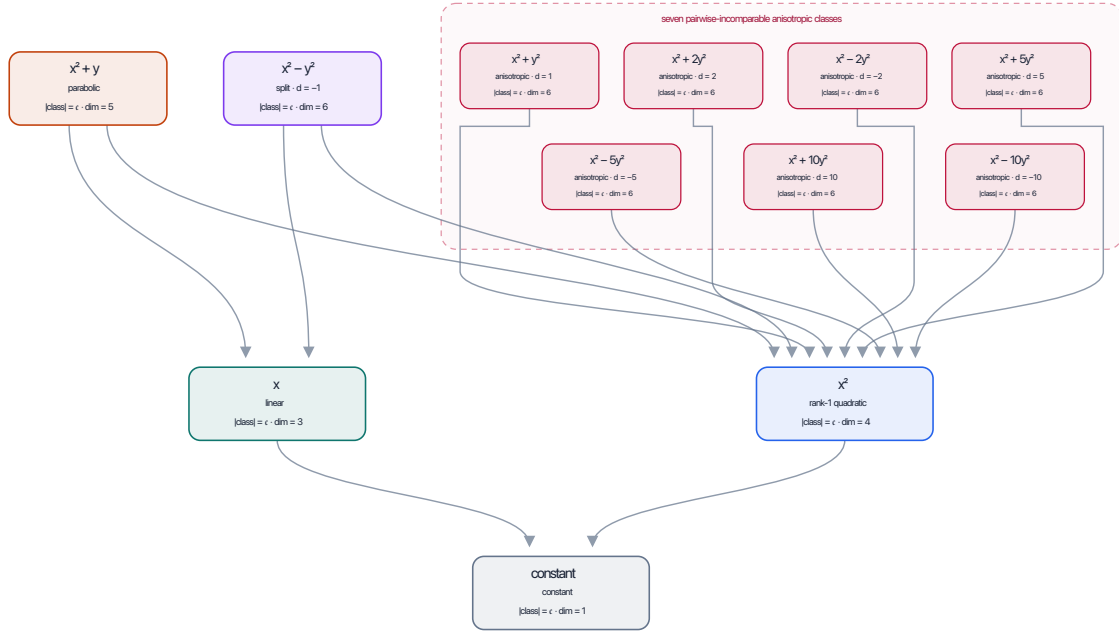


Figure 4: Quadratic maps $\mathbb{Q}_2^2 \rightarrow \mathbb{Q}_2$ with singular affine processors: twelve classes and thirteen covers. Seven anisotropic classes sit above the rank-one node. Their one-shot incomparability is exact; their block-affine exchange rates are largely unknown.

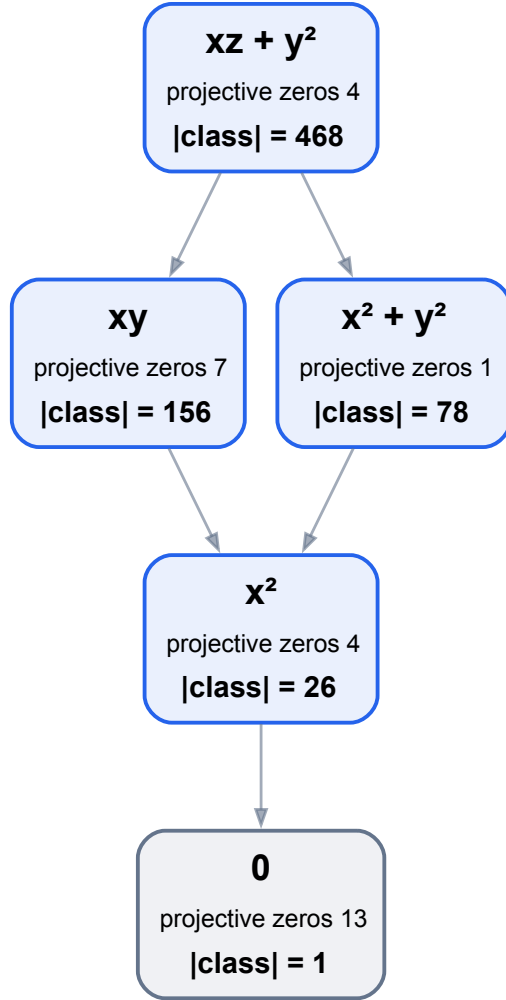


Figure 5: Ternary quadratic forms over $\mathbb{F}_3$, read as sections of $\mathcal{O}(2)$ on $\mathbf{P}^2$: five classes and five covers under possibly singular linear input processors and scalar output processors. The tensor classes are also five types of projective conic, including degenerate types.

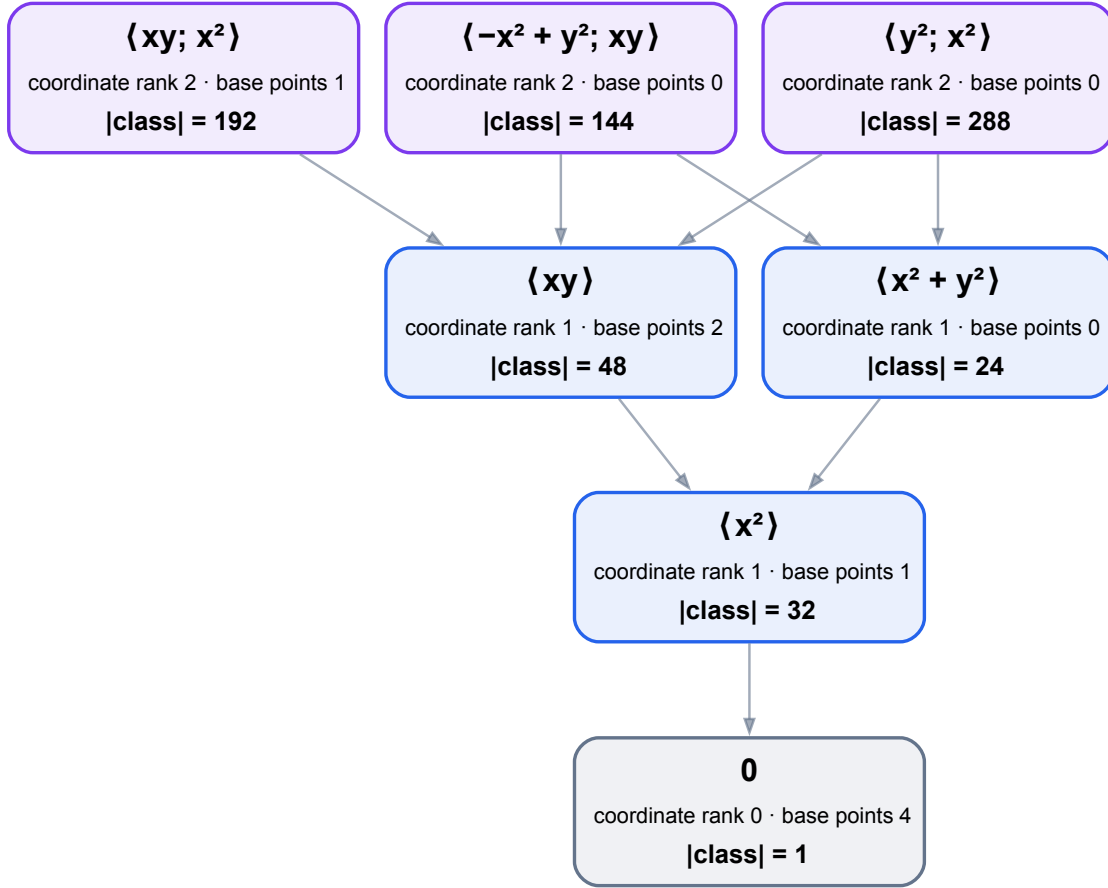


Figure 6: Seven homogeneous quadratic tensor classes and eight covers for $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$, with possibly singular linear input and output processors. This example will also carry a fiber-signature exchange matrix in Section 7.

For degree-at-most-three functions $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$, affine input and output processors give 14 classes and 22 Hasse covers. If quadratic input processors are allowed and the generated preorder is used, the 6,561 functions collapse to three mutual-reachability classes: surjective, two-valued with fibers $6 + 3$, and constant.

Over $\mathbb{F}_8$, the functional identities responsible for this collapse no longer hold. Among $8^{10} = 1,073,741,824$ cubic functions, the quadratic-input generated preorder has 110 classes. The exact Hasse diagram has 23 boxes after grouping order-theoretic twin antichains. This contrast is a warning: bounded-degree substitutions interact strongly with the arithmetic of the field and with polynomial identities as functions.

7 The fiber-signature observer

For a finite map $f : X \rightarrow Y$, discard the labels and retain the nonzero fiber sizes

$$\sigma(f) = (|f^{-1}(y_1)|, \dots, |f^{-1}(y_r)|).$$

The companion paper [1] studies a processor convention that injects each target fiber into an assigned source fiber. For that resource theory the exchange rate is

$$C_{\text{sig}}(g \rightarrow f) = \inf_{\beta \in [0, \infty]} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \quad Z_f(\beta) = \sum_y |f^{-1}(y)|^\beta. \quad (9)$$

Cartesian products multiply the partition functions, and the functions $Z_f(\beta)$ form the relevant asymptotic spectrum. Equivalently, (9) is homothetic containment of Gibbs energy–entropy regions.

Remark 7.1 (No silent identification with the affine rate). The affine and signature processors are not the same. An affine input may be singular, while the signature model preserves fibers by injections; conversely, the signature model forgets all algebraic restrictions. Therefore C_{sig} is used here as a computable observer attached to a polynomial map, not as a proved value or bound for C_{aff} . Any comparison theorem between the two rates requires an explicit functor between the processor categories.

7.1 A six-node matrix

Remove the zero node from Figure 6 and order the other nodes as

$$\begin{aligned} A_1 &= \langle x^2 \rangle, & A_2 &= \langle xy \rangle, & A_3 &= \langle x^2 + y^2 \rangle, \\ A_4 &= \langle xy; x^2 \rangle, & A_5 &= \langle y^2; x^2 \rangle, & A_6 &= \langle -x^2 + y^2; xy \rangle. \end{aligned}$$

Their signatures are

$$(6, 3), (5, 2, 2), (4, 4, 1), (3, 2, 2, 2), (4, 2, 2, 1), (2, 2, 2, 2, 1).$$

Thus the partition profiles for all seven nodes in Figure 6, including the zero node A_0 , are

$$\begin{aligned} \mathcal{Z}_0(s) &= 9^s, \\ \mathcal{Z}_1(s) &= 6^s + 3^s, \\ \mathcal{Z}_2(s) &= 5^s + 2 \cdot 2^s, \\ \mathcal{Z}_3(s) &= 2 \cdot 4^s + 1, \\ \mathcal{Z}_4(s) &= 3^s + 3 \cdot 2^s, \\ \mathcal{Z}_5(s) &= 4^s + 2 \cdot 2^s + 1, \\ \mathcal{Z}_6(s) &= 4 \cdot 2^s + 1. \end{aligned}$$

Equation (9) gives the matrix whose rows are resources and columns are targets:

$$M_{\text{sig}} = \begin{pmatrix} 1 & .630930 & .630930 & .500000 & .500000 & .430677 \\ .897378 & 1 & .996600 & .792481 & .792481 & .682606 \\ .773706 & .861353 & 1 & .792481 & .792481 & .682606 \\ .613147 & .682606 & .755027 & 1 & .792481 & .861353 \\ .773706 & .861353 & .904077 & .985905 & 1 & .861353 \\ .386853 & .430677 & .500000 & .630930 & .500000 & 1 \end{pmatrix}.$$

Every entry is positive and every two-way product is at most one. More strongly, the full composition inequalities $M_{ij}M_{jk} \leq M_{ik}$ hold, as guaranteed by the operational signature theory.

The matrix is not symmetric. Its ordinary eigenvalues are approximately

$$4.55109, 0.70981, 0.35523, 0.21839, 0.08274 \pm 0.00412i.$$

The leading eigenvalue exceeding one does not represent arbitrage: ordinary matrix multiplication sums over many alternative paths, whereas a conversion chooses and composes a path. The relevant algebra for best-path conversion is max-times, not ordinary linear algebra.

8 Cycle means: what they measure and what they do not

For a directed simple cycle $\mathcal{C}$, define its total return and per-step retention by

$$R(\mathcal{C}) = \prod_{(i,j) \in \mathcal{C}} M_{ij}, \quad r(\mathcal{C}) = R(\mathcal{C})^{1/|\mathcal{C}|}.$$

The no-arbitrage theorem gives $r(\mathcal{C}) \leq 1$. In logarithmic costs, $-\log r(\mathcal{C})$ is the mean loss per conversion. Maximizing this geometric mean is the maximum-mean-cycle problem, or the max-times spectral radius after trivial diagonal loops are removed. It is computable by the additive cycle-mean methods associated with [13].

For the six-node matrix above, the best simple cycle of each length is:

length	$r(\mathcal{C})$	$R(\mathcal{C})$	maximizing cycle
2	0.926512	0.858424	$A_2 \rightarrow A_3 \rightarrow A_2$
3	0.879489	0.680285	$A_2 \rightarrow A_3 \rightarrow A_5 \rightarrow A_2$
4	0.856880	0.539113	$A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow A_2$
5	0.789265	0.306278	$A_1 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow A_2 \rightarrow A_1$
6	0.741674	0.166448	$A_1 \rightarrow A_3 \rightarrow A_5 \rightarrow A_6 \rightarrow A_4 \rightarrow A_2 \rightarrow A_1$

These numbers say how nearly reversible the best closed portfolios are at a fixed number of distinct classes. They do not recover the Hasse order: a one-shot incomparable pair may still have positive block exchange rates.

For the seven pairwise-incomparable anisotropic $\mathbb{Q}_2$ classes, the true affine matrix is not known. A finite diagnostic reduces the forms modulo 2^{10} , forgets the affine structure, and computes their residue fiber-signature matrix. It has nontrivial cycles with mean exactly one among classes whose residue signatures coincide; its best full seven-class cycle has

$$r_7 = 0.9923402268, \quad R_7 = 0.9475980924.$$

This reveals that the residue observer is too coarse to retain the seven square classes. It is not evidence that the exact affine resources are reversible.

Thus cycle means have three defensible uses:

1. checking numerical no-arbitrage and composition consistency;
2. quantifying round-trip loss inside a chosen rate model;
3. detecting when an observer collapses algebraically distinct classes.

They do not by themselves supply a scalar complexity, a Hasse order, a positive-semidefinite form, or an affine coding theorem.

9 A certified fractional affine rate

Every affine rate quoted so far has been either an interval or a signature-observer value, which Remark 7.1 keeps separate from C_{aff} . This section proves an exact operational value strictly between zero and one for a pair of nondegenerate binary forms, and gives an explicit conversion in the opposite direction that is strictly better. The pair is

$$g_+(x, y) = x^2 + y^2, \quad g_-(x, y) = x^2 - y^2.$$

Over $\mathbb{R}$ these are the definite and the indefinite class of Section 5; over $\mathbb{F}_q$ with $q \equiv 3 \pmod{4}$ they are the anisotropic and the split class of Section 4. Both have rank two, so the rank monotone separates neither direction from rate one, and the content of the section is which finer monotone does.

9.1 Block implementation is a span condition

Lemma 9.1 (Homogeneous necessary condition). *Let $f, g: K^2 \rightarrow K$ be quadratic forms with symmetric matrices F, G , and let N_j denote the matrix of the j -th target block $f(z_{2j-1}, z_{2j})$, viewed as a form on K^{2k} . If $f^{\times k} \preceq_{\text{aff}} g^{\times r}$, then there are matrices $A_i \in \text{Mat}_{2 \times 2k}(K)$ and scalars B_{ji} with*

$$N_j = \sum_{i=1}^r B_{ji} M_i, \quad M_i = A_i^T G A_i, \quad 1 \leq j \leq k. \quad (10)$$

Proof. Write the input processor as $a(z) = (A_1 z + t_1, \dots, A_r z + t_r)$ and the output processor as $b(u) = Bu + u_0$. Then

$$g(A_i z + t_i) = z^T M_i z + (\text{linear in } z) + (\text{constant}),$$

so comparing the degree-two homogeneous parts of $f^{\times k} = b \circ g^{\times r} \circ a$ gives (10). Translations move only the lower-degree parts, so the condition is necessary whether or not the processors are linear. $\square$

Every target block must therefore lie in the span of the r pullbacks. The rate question becomes: how many pullbacks does such a span cost?

Lemma 9.2 (Pullback monotones). *Let $M = A^T G A$. Then $\text{rank } M \leq \text{rank } G$ over any field, and over $\mathbb{R}$ the inertia indices satisfy $n_{\pm}(M) \leq n_{\pm}(G)$.*

Proof. The rank statement is standard. If S is a subspace on which M is positive definite, then A is injective on S , because $Az = 0$ forces $z^\top Mz = 0$, and G is positive definite on $A(S)$. Hence $n_+(M) \leq n_+(G)$, and n_- follows by negating. $\square$

Proposition 9.3 (Additive rate bounds). *Let f, g be nonzero binary quadratic forms. Then*

$$C_{\text{aff}}(g \rightarrow f) \leq \frac{\text{rank } g}{\text{rank } f}, \quad (11)$$

and over $\mathbb{R}$,

$$C_{\text{aff}}(g \rightarrow f) \leq \frac{\max(n_+(g), n_-(g))}{\max(n_+(f), n_-(f))}. \quad (12)$$

Proof. Sum (10) over j . The left-hand side becomes the orthogonal sum of k copies of f on K^{2k} , of rank $k \text{rank } f$ and, over $\mathbb{R}$, of inertia $(kn_+(f), kn_-(f))$. The right-hand side is $\sum_i d_i M_i$ with $d_i = \sum_j B_{ji}$. Rank is subadditive under matrix addition, and so is n_+ : if $X + Y$ is positive definite on S while $X \preceq 0$ on K_X and $Y \preceq 0$ on K_Y , then $S \cap K_X \cap K_Y = 0$, whence $\dim S \leq n_+(X) + n_+(Y)$. Each summand obeys $\text{rank}(d_i M_i) \leq \text{rank } G$ and $n_+(d_i M_i) \leq \max(n_+(M_i), n_-(M_i))$, the second because a negative d_i exchanges the two indices; Lemma 9.2 then bounds both by the corresponding invariant of g . Therefore $k \text{rank } f \leq r \text{rank } g$ and $k \max(n_+(f), n_-(f)) \leq r \max(n_+(g), n_-(g))$ for every admissible pair (k, r) , and (4) passes to the limit. $\square$

9.2 The hyperbolic form pays double

Theorem 9.4 (An exact fractional rate). *Over $\mathbb{R}$,*

$$C_{\text{aff}}(x^2 - y^2 \rightarrow x^2 + y^2) = \frac{1}{2}.$$

Proof. For the upper bound apply (12): the source has inertia $(1, 1)$ and the target $(2, 0)$, so $C_{\text{aff}} \leq 1/2$. For the lower bound, two source copies restricted to $(x, 0)$ and $(0, y)$ produce x^2 and $-y^2$; the output processor $u_1 - u_2$ returns $x^2 + y^2$. Hence $k_{g_- \rightarrow g_+}(2) \geq 1$ and $C_{\text{aff}} \geq 1/2$. $\square$

The rank bound (11) gives only $C_{\text{aff}} \leq 1$ here. The fractional value is produced by the inertia refinement, which exists over $\mathbb{R}$ and has no counterpart over $\mathbb{F}_q$ or $\mathbb{Q}_p$; this is exactly the gap that leaves the pair of Section 10 open.

9.3 Three copies for two in the other direction

Proposition 9.5 (A three-to-two certificate). *Let the three input processors read the coordinate pairs (y_1, y_2) , (x_1, x_2) , (x_1, y_2) of $z = (x_1, y_1, x_2, y_2)$ and let the output processor be $B = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$. Then*

$$\begin{aligned} (x_1^2 + y_2^2) - (y_1^2 + y_2^2) &= x_1^2 - y_1^2, \\ (x_1^2 + x_2^2) - (x_1^2 + y_2^2) &= x_2^2 - y_2^2, \end{aligned} \quad (13)$$

so $g_-^{\times 2} \preceq_{\text{aff}} g_+^{\times 3}$ over every commutative ring and

$$C_{\text{aff}}(g_+ \rightarrow g_-) \geq \frac{2}{3}.$$

The processors are linear, so (13) is a single polynomial identity, valid over $\mathbb{R}$, over $\mathbb{F}_q$, and over $\mathbb{Q}_p$ alike. Its shape explains the saving. Read each of the three pullbacks as the unordered pair of coordinates it uses; the three pairs form the path

$$y_1 - y_2 - x_1 - x_2,$$

a spanning tree of the four coordinates in which the two coordinates of each target block sit at even distance. Even distance is what lets an alternating difference of the tree edges cancel one coordinate and reverse the sign of another, which is precisely what turns definite pullbacks into hyperbolic blocks. A tree on the $2m$ coordinates of m blocks has $2m - 1$ edges, and a single block cannot stand alone, since its own edge would put its two coordinates at odd distance. Splitting k blocks into pairs therefore costs $3k/2$ copies, and no other component size is cheaper, because $(2m - 1)/m$ is smallest at $m = 2$. This is what the construction offers; whether the span condition itself is that expensive is the open part.

Corollary 9.6 (A certified one-way pair). *Over $\mathbb{R}$,*

$$\frac{1}{2} = C_{\text{aff}}(g_- \rightarrow g_+) < \frac{2}{3} \leq C_{\text{aff}}(g_+ \rightarrow g_-) \leq 1,$$

so the two classes are asymptotically interconvertible at positive rates in both directions while the round trip loses at least a factor two:

$$\frac{1}{3} \leq R = C_{\text{aff}}(g_+ \rightarrow g_-) C_{\text{aff}}(g_- \rightarrow g_+) \leq \frac{1}{2}.$$

This is the first pair of nondegenerate binary classes in this paper with a proved strictly fractional affine rate, and the first certified strict asymmetry between two classes that the one-shot poset of Section 5 leaves incomparable. Compare Section 8: here the cycle mean bound is not an observer artifact but a proved property of the affine resources.

9.4 What the searches say

Neither (11) nor (12) bounds $C_{\text{aff}}(g_+ \rightarrow g_-)$ below one, so the exact value in that direction is not settled by the monotones above. Two searches probe it. Both test the span condition (10) directly, and neither is promoted to a proof.

Over $\mathbb{R}$, a multistart minimization of the distance from the target blocks to the span of r pullbacks gives residuals that vanish exactly at the block sizes the constructions reach:

source	target	k	r	span condition
definite	hyperbolic	1	1	no solution found
definite	hyperbolic	1	2	solved
definite	hyperbolic	2	2	no solution found
definite	hyperbolic	2	3	solved
definite	hyperbolic	3	4	no solution found
definite	hyperbolic	4	5	no solution found
definite	hyperbolic	5	7	no solution found
definite	hyperbolic	4	6	solved
hyperbolic	definite	1	1	no solution found
hyperbolic	definite	1	2	solved
hyperbolic	definite	2	3	no solution found
hyperbolic	definite	2	4	solved
hyperbolic	definite	3	5	no solution found
hyperbolic	definite	3	6	solved

The hyperbolic source is solved exactly at $k/r = 1/2$, as Theorem 9.4 requires. The definite source is solved at $2/3$ and at nothing larger: the three probes with $k/r > 2/3$ fail, including $k/r = 5/7$, which the paired-tree construction does not reach.

Over $\mathbb{F}_3$, where g_+ is anisotropic and g_- is split, the same question is finite and was settled exhaustively for $r = k + 1$:

source	target	k	r	pullbacks	conversion
anisotropic	split	2	3	861	exists
anisotropic	split	3	4	66,795	none
split	anisotropic	2	3	1,641	none
split	anisotropic	3	4	132,861	none

The $k = 2$ row recovers (13), now by exhaustion rather than by hand; the $k = 3$ rows exclude the rate $3/4$ at that block size in both directions.

Remark 9.7 (Why the finite search is exhaustive). With $r = k + 1$ the span W of the pullbacks has dimension at most $k + 1$ and contains the k -dimensional span V of the target blocks. The program first verifies that V contains no nonzero pullback, which excludes $W = V$; hence $W = V \oplus \langle P \rangle$ for every generator P . Over $\mathbb{F}_q$ a nonzero multiple of a pullback is again a pullback, since a binary form is determined by its rank and discriminant class, so each generator normalizes to $P + v$ with $v \in V$ without changing W . The search then only has to ask whether some pullback P has $\{v \in V : P + v \text{ is a pullback}\}$ spanning V , and it enumerates every pullback.

Both searches are consistent with $C_{\text{aff}}(g_+ \rightarrow g_-) = 2/3$, over $\mathbb{R}$ and over $\mathbb{F}_3$ alike, and neither proves it. An upper bound below one has to leave the additive monotones of Proposition 9.3 behind, since rank and inertia are both already spent in that direction. This is Question 15.2.

Regenerate the certificate check and both searches with `analysis/definite_hyperbolic_affine_rates.py`.

10 The unresolved 2-adic affine rates

Let

$$q_d(x, y) = x^2 + dy^2.$$

For the pair $q_1 = x^2 + y^2$ and $q_2 = x^2 + 2y^2$, two copies of either form implement one copy of the other: restrict two source blocks to $(x, 0)$ and $(y, 0)$, then combine the outputs. Therefore

$$C_{\text{aff}}(q_1 \rightarrow q_2), C_{\text{aff}}(q_2 \rightarrow q_1) \geq \frac{1}{2}.$$

An equal-block n -to- n conversion is impossible for distinct determinant square classes, but this does not exclude rates approaching one. What is missing here is the monotone that settles the real pair of Section 9: over $\mathbb{Q}_2$ there is no inertia, and rank and discriminant stop at rate one. The current rigorous bounds remain

$$\frac{1}{2} \leq C_{\text{aff}}(q_1 \rightarrow q_2) \leq 1, \quad \frac{1}{2} \leq C_{\text{aff}}(q_2 \rightarrow q_1) \leq 1.$$

No integral-lattice 3-to-2 identity exists modulo four, and a natural shared-summand ansatz also fails. Rational processors with denominators and general cross-copy mixing remain open.

There is a separate local-zeta diagnostic. With normalized Haar measure on $\mathbb{Z}_p^n$, define

$$Z_{f,p}(s) = \int_{\mathbb{Z}_p^n} |f(x)|_p^s dx, \quad C_\zeta(f \rightarrow g) = \inf_{s>0} \frac{-\log Z_{f,p}(s)}{-\log Z_{g,p}(s)}.$$

For $t = 2^{-s}$,

$$Z_{x^2,2}(s) = \frac{1}{2-t^2}, \quad Z_{x^2-y^2,2}(s) = \frac{t^2-2t+2}{(2-t)^2},$$

giving

$$C_\zeta(x^2 \rightarrow x^2 - y^2) = 0.9397027875 \dots$$

Igusa's rationality theory explains why these local profiles are rational in p^{-s} [14]. But they depend on the chosen lattice and are invariant under $\text{GL}_n(\mathbb{Z}_p)$, not unrestricted $\text{GL}_n(\mathbb{Q}_p)$. They are prices in a local profile model, not the operational affine rates above.

11 Fibrations of the affine plane over $\mathbb{F}_q$

Section 7 evaluated the fiber-signature observer on orbit classes of quadratic maps over $\mathbb{F}_3$, where the signatures are short and carry no arithmetic. This section evaluates the same observer on the smallest family of maps whose signatures are arithmetically rich: polynomials $f \in \mathbb{F}_q[x, y]$, read as maps $\mathbb{A}^2 \rightarrow \mathbb{A}^1$, whose fibers are plane curves with a genus and a point count. Both directed rates against a linear map then have closed forms, and each of them reads exactly one classical statistic of the family.

Subsection 11.4 states, without softening, what that does and does not amount to. A reader who wants the objection before the results should read it first.

11.1 Setup and notation

Fix an odd prime power q and $f \in \mathbb{F}_q[x, y]$. Write

$$N_c = \#f^{-1}(c) = \#\{(x, y) \in \mathbb{F}_q^2 : f(x, y) = c\}, \quad a_c = q - N_c,$$

so $\sigma(f) = \{N_c\}_{c \in \mathbb{F}_q}$ is the fiber signature of Section 7 and a_c is the deviation of the fiber count from the generic value q . When the fiber over c is a smooth affine curve of genus g with one rational

point at infinity, a_c is the trace of Frobenius of its smooth projective model and $|a_c| \leq 2g\sqrt{q}$ by the Riemann hypothesis for curves [9]. Let

$$L = (q, q, \dots, q) \quad (q \text{ entries})$$

be the flat signature of the linear map $f(x, y) = x$. It is the natural reference: same number of fibers and same total q^2 as any f , but perfectly even.

Two elementary identities govern everything below. First, counting the plane twice,

$$\sum_{c \in \mathbb{F}_q} N_c = q^2 \quad \implies \quad \sum_{c \in \mathbb{F}_q} a_c = 0 \quad (14)$$

identically, for every f , the sums being over all of $\mathbb{F}_q$. The signature retains only the nonzero fiber sizes, so it determines $\{a_c\}$ exactly when the image is full, which holds for every family considered below. Second, with

$$m_2 = \frac{1}{q^2} \sum_{c \in \mathbb{F}_q} a_c^2, \quad (15)$$

the Weil bound gives $m_2 = O(g^2)$, and for a family with large geometric monodromy the normalized traces $a_c/\sqrt{q}$ equidistribute for the trace law of $USp(2g)$ [10], so $m_2 \rightarrow 1$. The cyclic family $y^2 = x^3 + c$ instead has $m_2 \rightarrow 2$ at $q \equiv 1 \pmod{3}$ and $m_2 = 0$ at $q \equiv 2 \pmod{3}$.

Example 11.1 ($f = y^2 - x^3 - x$ at $q = 11$). Listing the fiber counts in decreasing order,

$$\begin{aligned} \sigma(f) &= (17, 15, 14, 13, 12, 11, 10, 9, 8, 7, 5), & \sum N_c &= 121 = q^2, \\ \{a_c\} &= (-6, -4, -3, -2, -1, 0, 1, 2, 3, 4, 6), & \sum a_c &= 0, \end{aligned}$$

and $\sum a_c^2 = 132$, so $m_2 = 132/121 = 1.0909\dots$

Remark 11.2 (Geometric irreducibility is a real hypothesis). Everything in this section that is stated at the Weil scale $1/\sqrt{q}$ assumes every fiber is geometrically irreducible over $\mathbb{F}_q$. A fiber with r geometric components carries $N \approx rq$ points, and then $1 - C_{\text{sig}}(L \rightarrow f) \asymp \log r / \log q$, which swamps every Weil-scale term. The split conic $f = xy$ has $N_0 = 2q - 1$ and is exactly this failure; the correct genus-zero control is $f = x^2 + y^2$ at $q \equiv 3 \pmod{4}$, where every nonzero fiber is an anisotropic conic with $a_c = -1$ and $1 - C_{\text{sig}}(L \rightarrow f) = 1/(q \log q)$. For $f = y^2 - h(x)$ with h squarefree of odd degree the singular fibers are nodal and irreducible, which is why the hyperelliptic families below behave.

11.2 The forward rate is an endpoint rate

Proposition 11.3 (Forward rate). *For every $f : \mathbb{A}^2 \rightarrow \mathbb{A}^1$ over $\mathbb{F}_q$ with full image and not all fibers equal,*

$$C_{\text{sig}}(L \rightarrow f) = \frac{\log q}{\log \max_c N_c},$$

and the infimum in (9) is attained at $\beta = \infty$.

Proof. Write $A = \log q$, $B = \log \max_c N_c$, and note $B > A$ because the N_c sum to q^2 over q fibers and are not all equal. Since $Z_L(\beta) = q^{1+\beta}$ and $Z_f(\beta) = \sum_c N_c^\beta \leq q(\max_c N_c)^\beta$,

$$\frac{\log Z_L(\beta)}{\log Z_f(\beta)} \geq \frac{(1+\beta)A}{A+\beta B} > \frac{A}{B}$$

for every finite β , the last inequality being exactly $B > A$; and the ratio tends to A/B as $\beta \rightarrow \infty$. $\square$

So this rate sees the largest fiber and nothing else. Verified against the solver of [1] to ten decimals:

q	family	$\max_c N_c$	$C_{\text{sig}}(L \rightarrow f)$	contact
211	$g = 1, y^2 = x^3 + x + c$	239	0.9772471017	$\beta = \infty$
211	$g = 2, y^2 = x^5 + x + c$	249	0.9699870998	$\beta = \infty$
211	$g = 3, y^2 = x^7 + x + c$	250	0.9692829874	$\beta = \infty$
1009	$g = 1, y^2 = x^3 + x + c$	1071	0.9914521108	$\beta = \infty$
1009	$g = 2, y^2 = x^5 + x + c$	1102	0.9874134968	$\beta = \infty$
1009	$g = 3, y^2 = x^7 + x + c$	1090	0.9889592944	$\beta = \infty$

Writing $\mu = \max_c(-a_c)/\sqrt{q}$, Proposition 11.3 gives the exact expansion

$$(1 - C_{\text{sig}}(L \rightarrow f))\sqrt{q} \log q = \mu - \frac{\mu^2}{\sqrt{q}} \left(\frac{1}{2} + \frac{1}{\log q} \right) + O(q^{-1}), \quad (16)$$

so the asymptotics of the forward rate are exactly the extreme-value statistics of the Frobenius traces.

Remark 11.4 (The $2g$ limit and the rate of approach; numerical). Under Katz–Sarnak equidistribution $\mu \rightarrow 2g$, but the approach is slow and its exponent is a finer invariant than g . The $USp(2g)$ trace law has lower tail $P(2g - T < \varepsilon) \sim K_g \varepsilon^{d/2}$ with $d = 2g^2 + g = \dim USp(2g)$ — the edge exponent is half the dimension of the monodromy group — whence a deficit $2g - \mu \approx \Gamma(1 + 2/d) (qK_g)^{-2/d}$. Log–log fits over $q \in [4 \cdot 10^3, 10^6]$ for four families per genus give:

g	fitted exponent	$-2/(2g^2 + g)$	μ at $q = 10^6$ (target $2g$)
2	−0.1969	−0.2000	3.684 (4)
3	−0.0931	−0.0952	4.663 (6)
4	−0.0583	−0.0556	4.760 (8)

The honest reading of the last column is that the $2g$ law is confirmed as an asymptotic and is numerically unreachable for $g \geq 3$: within 10% of $2g$ requires $q \approx 6 \cdot 10^4, 6 \cdot 10^9, 4 \cdot 10^{16}$ for $g = 2, 3, 4$. The reason is that the $USp(2g)$ trace has variance 1 and is nearly Gaussian in the bulk for every g , so the maximum of q samples sits near $\sqrt{2 \log q} = 4.77$ at $q = 10^6$ irrespective of the genus; the Weil edge is felt only once $\log q \gtrsim 2g^2$. At $g = 4$ the observed maximum agrees with the Gaussian prediction to three decimals and $2g = 8$ plays no role at all. For $g = 1$ the deficit is not extreme-value but integer quantization: $\max_c(-a_c) = \lfloor 2\sqrt{q} \rfloor$ for essentially every $q \gtrsim 4000$.

11.3 The reverse rate has a universal bottleneck temperature

Proposition 11.5 (Reverse rate). *Let f have full image and $\max_c |a_c| = O(\sqrt{q})$. Then*

$$1 - C_{\text{sig}}(f \rightarrow L) = \frac{m_2}{2q \log q} \max_{\beta \geq 0} \frac{\beta(1 - \beta)}{\beta + 1} + O(q^{-3/2}),$$

and the maximum is attained at the positive root of $\beta^2 + 2\beta - 1 = 0$:

$$\beta^* = \sqrt{2} - 1 = 0.414213562\dots, \quad 1 - C_{\text{sig}}(f \rightarrow L) = \frac{(3 - 2\sqrt{2})m_2}{2q \log q} + O(q^{-3/2}). \quad (17)$$

Both constants are independent of the family, the genus and q .

Derivation. Expand $N_c = q(1 - a_c/q)$, so that

$$\log Z_f(\beta) = (\beta + 1) \log q + \sum_{k \geq 2} (-1)^k \binom{\beta}{k} m_k q^{-k/2} + O(q^{-2}), \quad m_k = q^{-1-k/2} \sum_c a_c^k.$$

The term $k = 1$ is absent by (14): this is the point at which the identity $\sum_c N_c = q^2$ enters. Dividing by $\log Z_L(\beta) = (\beta + 1) \log q$ gives

$$\frac{\log Z_f(\beta)}{\log Z_L(\beta)} = 1 + \frac{\binom{\beta}{2} m_2}{(\beta + 1) q \log q} + O(q^{-3/2}),$$

so, taking the infimum over β and keeping the leading term,

$$1 - C_{\text{sig}}(f \rightarrow L) = \frac{m_2}{q \log q} \max_{\beta \geq 0} \frac{-\binom{\beta}{2}}{\beta + 1} + O(q^{-3/2}), \quad \binom{\beta}{2} = \frac{\beta(\beta - 1)}{2}.$$

Since $\binom{\beta}{2} < 0$ exactly on $0 < \beta < 1$, the infimum is interior. Differentiating $\beta(1 - \beta)/(\beta + 1)$ gives $-(\beta^2 + 2\beta - 1)/(\beta + 1)^2$, whose positive root is $\sqrt{2} - 1$; the value there is $3 - 2\sqrt{2}$. $\square$

Measured against the solver at $q = 1009$, the observed contact temperature and the observed constant $(1 - C_{\text{sig}}) \cdot 2q \log q / m_2$ are:

q	family	observed β^*	observed constant
1009	$g = 1, y^2 = x^3 + x + c$	0.414089	0.171802
1009	$g = 2, y^2 = x^5 + x + c$	0.414128	0.171671
1009	$g = 3, y^2 = x^7 + x + c$	0.413947	0.172101
	targets	0.414214	0.171573

For $f = y^2 - x^3 - x$ the leading term of (17) against the true value is $3.548 \cdot 10^{-3}$ versus $3.869 \cdot 10^{-3}$ at $q = 11$ (a 9% discrepancy, the dropped term being of order $q^{-3/2} \approx 0.027$), and $1.226 \cdot 10^{-5}$ versus $1.227 \cdot 10^{-5}$ at $q = 1009$.

The interior is not exhausted by m_2 . The same expansion gives $1 - C_{\text{sig}}(f \rightarrow L) = \sum_{k \geq 2} c_k m_k + O(q^{-2})$ with successive ratio

$$\frac{c_{k+1}}{c_k} = -\frac{\beta^* - k}{(k + 1)\sqrt{q}} = \frac{0.5286}{\sqrt{q}}, \frac{0.6464}{\sqrt{q}}, \frac{0.7172}{\sqrt{q}}, \dots \quad (18)$$

so the whole moment ladder is present, each order damped by a further $\approx 0.6/\sqrt{q}$. Subtracting the predicted contributions of moments $2, \dots, K$ from $1 - C_{\text{sig}}(f \rightarrow L)$ over about 720 hyperelliptic maps of genus 1–6 at $q = 503$ drops the residual root-mean-square from $2.75 \cdot 10^{-5}$ to $8.5 \cdot 10^{-8}$, $5.7 \cdot 10^{-8}$, $3.7 \cdot 10^{-9}$, after which it plateaus at exactly the first dropped nonlinear term (predicted $3.30 \cdot 10^{-9}$, observed $3.36 \cdot 10^{-9}$).

11.4 The direction of information flow

The two propositions must now be read against an objection that any number theorist will raise on sight, and that we state in its strongest form.

Remark 11.6 (The reduction, stated openly). $C_{\text{sig}}(L \rightarrow f)$ is a strictly monotone function of $\max_c N_c$, and $1 - C_{\text{sig}}(f \rightarrow L)$ is, to leading order, a linear function of m_2 . So the sentence “the exchange rate encodes the extreme Frobenius trace and the second moment” reduces on inspection to “a monotone function of X is a function of X ”. More generally the rate is a function of the signature, and the signature is nothing but the list $\{N_c\}$; therefore *nothing the rate says about the arithmetic of f could fail to be computable from $\{N_c\}$ directly*. Information flows from the arithmetic into the matrix, not out of it. No result in this section is a new theorem about curves over finite fields, and none should be advertised as one.

What survives Remark 11.6 is smaller, and worth stating exactly.

1. **The compression is selected, not chosen.** There are many ways to compress $\{N_c\}$ into a real number. The point of Propositions 11.3 and 11.5 is not that $\max_c N_c$ and m_2 are readable — they are readable by definition — but that these two, and no others, are what the *resource-theoretic comparison against a linear map* returns. The framework was not built with Frobenius traces in view, and the statistics it selects are exactly the two that the arithmetic theory already regards as fundamental. That is a consistency check on the framework, not a discovery about curves.
2. **The bottleneck constant is a fact about the comparison, and carries no arithmetic.** $\beta^* = \sqrt{2} - 1$ is independent of family, genus and q ; it is where the Gibbs curve of any Hasse–Weil-normalized signature makes contact with the flat one. Its content is structural: the reverse comparison of a curve fibration with a linear map is decided at one fixed finite temperature, never at an endpoint, and the vanishing of the first moment (14) is what puts it there.
3. **The negative statements are genuine statements about the framework.** Which statistics the rate provably *cannot* see is not computable from $\{N_c\}$ by inspection, because it is a statement about the comparison rather than about f . Subsection 11.6 collects them.
4. **The one route that would escape the reduction is open.** A scalar reading of a family is a reading only because $\preceq$ happens to be a total preorder on the families compared. The comparison $a \preceq b \iff C_{\text{sig}}(a \rightarrow b) < C_{\text{sig}}(b \rightarrow a)$ is known to run in circles for integer signatures and for homogeneous maps $\mathbb{F}_3^3 \rightarrow \mathbb{F}_3^3$ [1]. If three families of curves over a fixed $\mathbb{F}_q$ cycled, then no real-valued invariant φ of such families could satisfy $a \preceq b \iff \varphi(a) < \varphi(b)$, and the comparison would not be a function of any single number — the one statement in this circle of ideas that is not a repackaging of a classical statistic. We have not found such a cycle, and Question 15.5 records why the present regime resists one: a pair whose two rates are both attained at an endpoint is ordered by the scalar $\varphi = \log(\#\text{fibers}) \cdot \log(\max \text{fiber})$ and therefore cannot sit on a cycle, and curve signatures are close to flat, so the interior correction available to overturn a φ -gap is only $O(1/(q \log q))$.

The rest of this section is written under Remark 11.6: the statements are translations, and their interest is that the translation is exact and that the dictionary is short.

11.5 Flatness is a permutation-polynomial condition

The one place where an entry of the exchange matrix is *equivalent* to a named arithmetic condition, rather than approximately sensitive to a statistic, is the degenerate case $C_{\text{sig}} = 1$.

Theorem 11.7 (Flatness). *Let q be an odd prime power and $P \in \mathbb{F}_q[x]$. For $f(x, y) = y^2 - P(x)$, $C_{\text{sig}}(f \rightarrow L) = C_{\text{sig}}(L \rightarrow f) = 1 \iff \sigma(f) = L \iff P$ is a permutation polynomial of $\mathbb{F}_q$.*

Proof. With χ the quadratic character, $a_c = -\sum_x \chi(P(x) + c)$, and orthogonality of χ gives the exact identity

$$\sum_c a_c^2 = q(\nu(P) - q), \quad \nu(P) = \#\{(x, x') \in \mathbb{F}_q^2 : P(x) = P(x')\}. \quad (19)$$

So $\sigma(f) = L$ iff $\nu(P) = q$. Writing $n_u = \#P^{-1}(u)$ we have $\sum_u n_u = q$ and $\sum_u n_u^2 = q$ with $n_u \geq 0$ integers, which forces $n_u \in \{0, 1\}$ and hence $n_u \equiv 1$. For the first equivalence, put $h(\beta) = \log Z_f(\beta) - \log Z_L(\beta)$. Since $\log Z_L > 0$ on $[0, \infty]$, $C_{\text{sig}}(f \rightarrow L) = 1$ forces $h \geq 0$ and $C_{\text{sig}}(L \rightarrow f) = 1$ forces $h \leq 0$; so both hold iff $h \equiv 0$, that is iff the Dirichlet series $\sum_c N_c^\beta$ and $\sum_c q^\beta$ agree identically, which happens iff the two multisets agree. (The shape of h is fixed in any case: it is convex, vanishes at $\beta = 1$ and is ≤ 0 at $\beta = 0$, hence ≤ 0 on $[0, 1]$ and ≥ 0 on $[1, \infty)$.) $\square$

Verified on 2093 pairs (P, q) — 23 branch polynomials against all 91 primes $11 \leq q \leq 500$ — with no exceptions. The instances are classical [7]:

branch map P	flat exactly when	note
x^d	$\gcd(d, q-1) = 1$	$d = 3$ recovers the $q \equiv 2 \pmod 3$ case
$D_n(x)$ (Dickson)	$\gcd(n, q^2-1) = 1$	involves $q+1$; D_5 flat iff $q \equiv \pm 2 \pmod 5$
$x^3 + ax$, $a = 1, 2$	never, for $q \leq 500$	negative control

The same argument in the other variable gives, for $f = y^r - x^d$, flatness exactly when $\min(\gcd(r, q-1), \gcd(d, q-1)) = 1$ — verified on 4368 triples $(r \leq 7, d \leq 9, q \leq 500)$ with no exceptions — so $C_{\text{sig}}(y^3 - x^5 \rightarrow L) = 1$ detects $q \not\equiv 1 \pmod{15}$. The Dickson row is the qualitatively different one: it is a condition on q modulo n through $q^2 - 1$, which no monomial family can produce.

Example 11.8 (A non-congruence entry). For a fixed $E : y^2 = P(x)$ put $f(x, y) = P(x)y^2$ and let $z = \#\{x : P(x) = 0\}$, $a_E = q - \#\{(x, y) : y^2 = P(x)\}$. Then exactly

$$N_0 = q + z(q-1), \quad N_c = q - z - \chi(c)a_E \quad (c \neq 0),$$

so the signature takes at most three values, and it equals that of the split conic xy precisely when $z = 1$ and $a_E = 0$. Also $z = 0$ forces $\#E(\mathbb{F}_q)$ odd, hence $a_E \neq 0$, so a quadratic-twist family is never flat. For the non-CM curve $y^2 = x^3 + x + 1$ the primes $q \leq 500$ with $a_E = 0$ are 17, 179, 227 — the supersingular primes, an infinite [11] set of density zero [12]. At 17 and 179 the branch cubic has a single root and the signature is exactly that of the split conic; at 227 it has three, and the signature is $(4q-3, (q-3)^{q-1})$. So the exchange-matrix entry $C_{\text{sig}}(f \rightarrow xy) = C_{\text{sig}}(xy \rightarrow f) = 1$ cuts out the supersingular primes with $z = 1$. That set is nonempty and of density zero, whereas by Dirichlet any nonempty union of admissible residue classes has positive density; so this entry detects a condition that provably is not a congruence, and not one the framework was designed to express.

Remark 11.9 (No exotic exact values). Scanning 69 426 ordered rates — all pairs from a 58-map structured pool at every prime $11 \leq q \leq 97$ — against every p/r with $r \leq 12$ and every $\log a / \log b$ with $a, b \leq 24$ at tolerance 10^{-12} , the only values that persist across q are 1 (signature coincidence), $1/2$ (which is $C_{\text{sig}}(f \rightarrow \text{const})$, a third equivalent form of flatness) and 0. The arithmetic enters the exchange matrix entirely through *which signatures coincide*, not through exotic rate values. The reason is structural: an exact value needs an endpoint infimum with commensurable logarithms, and for maps $\mathbb{A}^2 \rightarrow \mathbb{A}^1$ the fiber counts are all $q + O(\sqrt{q})$ or small multiples of q .

11.6 What the rate cannot see

Two invisibilities are structural rather than numerical, and they are the statements of this section that do not follow from inspecting $\{N_c\}$.

Proposition 11.10 (The smallest fiber is not an endpoint). *The convention $\beta \in [0, \infty]$ of (9) isolates $\max_c N_c$ at $\beta = \infty$ and $\#\{c : N_c > 0\}$ at $\beta = 0$, and has no endpoint isolating $\min_c N_c$; isolating it would require $\beta < 0$, which is excluded because $Z_f(\beta)$ is order-preserving in the signature only for $\beta \geq 0$. Consequently $\min_c a_c$ is recovered exactly, as an integer, while $\max_c a_c$ is seen only through the $q^{-k/2}$ -damped moments of (18).*

The asymmetry is sharp, and the witness also shows that the interior tangency is not redundant. Among 1600 full-image hyperelliptic maps at $q = 211$, grouped by the two endpoint readouts, the closest admissible pair is

$$\begin{aligned} f : P &= (151, 33, 27, 120, 193, 35, 61, 108, 199, 1), & N_{\min} &= 173, \\ g : P &= (200, 202, 15, 180, 196, 178, 98, 39, 4, 1), & N_{\min} &= 167, \end{aligned}$$

with equal image size, equal $N_{\max} = 248$ and *exactly* equal $m_2 = 0.966824644550$. All four endpoint probes agree exactly, to the last bit; the interior probes separate by $2.419 \cdot 10^{-7}$, against a solver resolution of 10^{-13} , and the separation is predicted in advance by the ladder (18) as $c_3 \Delta m_3 + c_4 \Delta m_4 = 2.32 \cdot 10^{-7}$ (agreement 4%). The six-unit change in the smallest fiber enters only through the moments.

Proposition 11.11 (The Katz–Sarnak separating statistic is unavailable). *In the Katz–Sarnak classification the low moment that separates orthogonal from symplectic symmetry is the first: $\int \text{tr} = 0$ for U , USp and $SO(\text{even})$ but ± 1 for the odd orthogonal cases, whereas $\int \text{tr}^2 = 1$ for symplectic and orthogonal alike [10]. By (14) the first moment of $\{a_c\}$ vanishes identically for every fibration of $\mathbb{A}^2$. So the statistic that would separate the symmetry types is identically zero for every resource in this class, and the next one available, m_2 , takes the same value for both; no exchange rate against any reference reads symmetry type at leading order at a single q .*

This is the same identity $\sum_c N_c = q^2$ that makes the framework work, so the obstruction is intrinsic rather than a limitation of the reference family chosen. Symmetry type reappears only as a statement about an ensemble: for a quadratic-twist family $m_2 = (1 - 1/q)a_E^2/q$ exactly, so $1 - C_{\text{sig}}(f \rightarrow L)$ fluctuates over q with the Sato–Tate law (over the 100 primes $q < 2000$ with $z = 0$ for $y^2 = x^3 + x + 1$, mean 0.844, standard deviation 0.901 against Sato–Tate values 1 and 1), whereas a large-monodromy family gives $m_2 = 0.9969 \pm 0.0035$ — deterministic.

What *is* visible beyond the genus is the monodromy rank. By (19), $m_2 = \nu(P)/q - 1$ exactly, and $\nu(P)/q$ converges to the number of orbits of the geometric monodromy group of P on ordered pairs. At $q = 601$:

P	monodromy	rank	m_2
x^3	cyclic $\mathbb{Z}/3$	3	1.9967
x^5	cyclic $\mathbb{Z}/5$	5	3.9933
D_5	dihedral	3	1.9834
random of degree 5	S_5	2	0.9617
random of degree 6	S_6	2	0.9750

Rank 2 means 2-transitive monodromy and gives $m_2 = 1$ at *every* genus, so m_2 is a monodromy invariant independent of the genus. At reachable q it is by far the sharper of the two readings: at $q = 421$ the m_2 reading is exact to three or four digits, while Remark 11.4 shows the genus reading does not resolve $g = 2, 3, 4$ at all. The two readings are also genuinely two-dimensional data: across a panel of 13 families they order the families differently in 10 of 13 positions at $q = 421$ and 12 of 13 at $q = 601$.

Remark 11.12 (Signature collisions are structural). Among 400 random elliptic fibrations $y^2 = P_3(x) + c$ there are only $\gcd(4, q - 1) + 1$ distinct signatures — 5 at $q = 101$, 3 at $q = 211, 503$ — because $x \mapsto u^2x, y \mapsto u^3y, c \mapsto u^6c$ shows the signature depends only on the quartic twist class. From genus 2 up, 398–400 of 400 random maps have distinct signatures. The rate cannot separate two members of one twist class, but that is a property of the signature, not a failure of the rate.

11.7 The partition function at integer temperature is a point count

One identity deserves separate mention because it connects this section to the next. For $f : X \rightarrow Y$ a map of finite sets and k a positive integer,

$$Z_f(k) = \sum_y (\#f^{-1}(y))^k = \#(X \times_Y X \times_Y \cdots \times_Y X) \quad (k \text{ factors}), \quad (20)$$

so the partition function of the signature observer, evaluated at an integer temperature, is the number of $\mathbb{F}_q$ -points of the k -fold fiber power. In particular, for $f : \mathbb{A}^2 \rightarrow \mathbb{A}^1$,

$$Z_f(2) = q^3 + q^2 m_2, \quad \text{that is} \quad m_2 = \frac{Z_f(2)}{q^2} - q,$$

so the second moment is not an auxiliary statistic attached to the rate: it is the partition function itself at $\beta = 2$. The fiber square $X \times_Y X$ is a threefold, and by Lang–Weil its leading term q^3 records that it has one geometric component; m_2 is exactly the normalized secondary term, which is the same place the Frobenius traces live. Thus for these resources the Gibbs variable β has a geometric meaning at the integers, and its extension to real β — which is what the infimum in (9) actually uses — interpolates between point counts of fiber powers. This is the closest the present framework comes to the zeta-function picture taken up in Section 12, and it is worth being precise about how close: it is an interpolation of counts, with no functional equation, no Euler product, and no zeros.

Remark 11.13 (Provenance). Propositions 11.3, 11.5, 11.10, 11.11 and Theorem 11.7 are proved above. The numbers in Remark 11.4, the moment-ladder residuals, the monodromy table and the collision census are computations, re-verified against the independent solver of [1] at a precision of 10^{-13} ; they are reported as computations. A grid truncated below $\beta \approx 500$ can hide interior structure on these objects entirely, so every contact temperature quoted here was confirmed on a dense grid to $\beta \sim 10^3$.

12 Prime modes and the zeta partition function

For $\Re s > 1$, unique factorization gives the exact identities

$$\zeta(s) = \sum_{n \geq 1} e^{-s \log n} = \prod_p \left(\sum_{k \geq 0} e^{-sk \log p} \right) = \prod_p \frac{1}{1 - p^{-s}}. \quad (21)$$

This is a genuine Gibbs interpretation for real $s = \beta > 1$. An “atom” at a prime p is an independent bosonic mode with occupation number $k \geq 0$ and energy

$$E_p(k) = k \log p.$$

Its partition function is

$$Z_p(\beta) = \frac{1}{1 - p^{-\beta}},$$

and the joint energy of an occupation vector $(k_p)_p$ is $\sum_p k_p \log p = \log n$. This elementary oscillator picture is compatible with, but much less structured than, the Bost–Connes quantum statistical system whose partition function is $\zeta(\beta)$ [15].

For one prime mode, the geometric occupation law gives

$$\bar{k}_p(\beta) = \frac{1}{p^\beta - 1}, \quad U_p(\beta) = \frac{\log p}{p^\beta - 1}, \quad S_p(\beta) = -\log(1 - p^{-\beta}) + \beta U_p(\beta).$$

Figure 7 compares the first six primes. At a fixed energy, a larger p has fewer accessible occupation levels and therefore smaller entropy. If $u = U_p / \log p = \bar{k}_p$, however, every curve reduces to the same universal bosonic relation

$$S = (1 + u) \log(1 + u) - u \log u.$$

Each individual prime mode is normalizable for every $\beta > 0$, although their infinite product is normalizable only for $\beta > 1$. No individual mode has a negative-temperature branch because its spectrum is unbounded above.

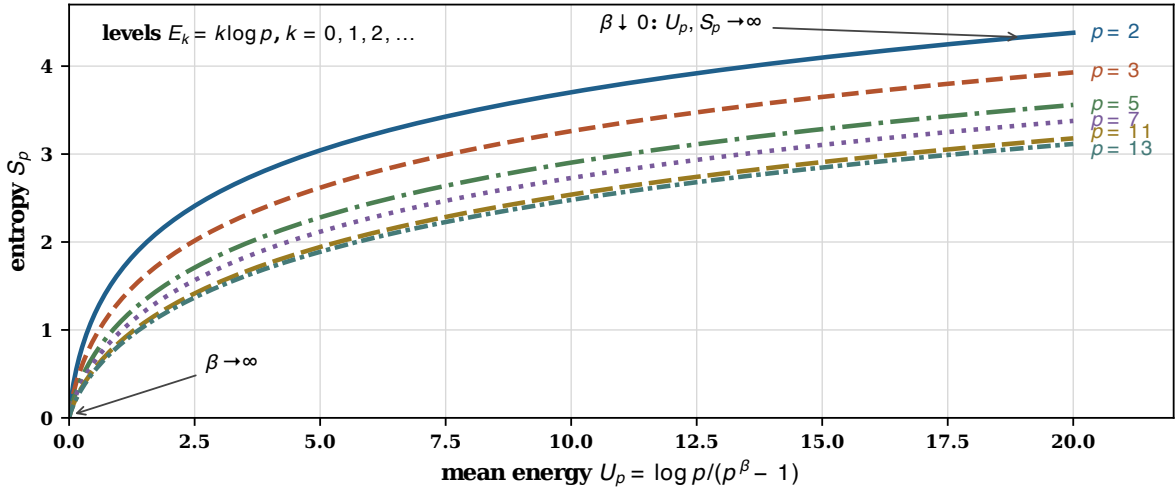


Figure 7: Entropy–energy curves for the first six prime modes $p = 2, 3, 5, 7, 11, 13$, each with levels $E_k = k \log p$. The curves are distinct in physical energy U_p , but coincide after the horizontal rescaling $U_p \mapsto U_p / \log p$. Their canonical domain is $\beta > 0$; negative temperatures do not define Gibbs states.

For the normalized Gibbs law, in units with $k_B = 1$, the mean energy and entropy are

$$U(\beta) = -\frac{d}{d\beta} \log \zeta(\beta), \quad S(\beta) = \log \zeta(\beta) + \beta U(\beta).$$

Figure 8 plots this parametric entropy–energy curve. It has only the branch $\beta > 1$: as $\beta \rightarrow \infty$, both U and S tend to zero, while both diverge as $\beta \downarrow 1$.

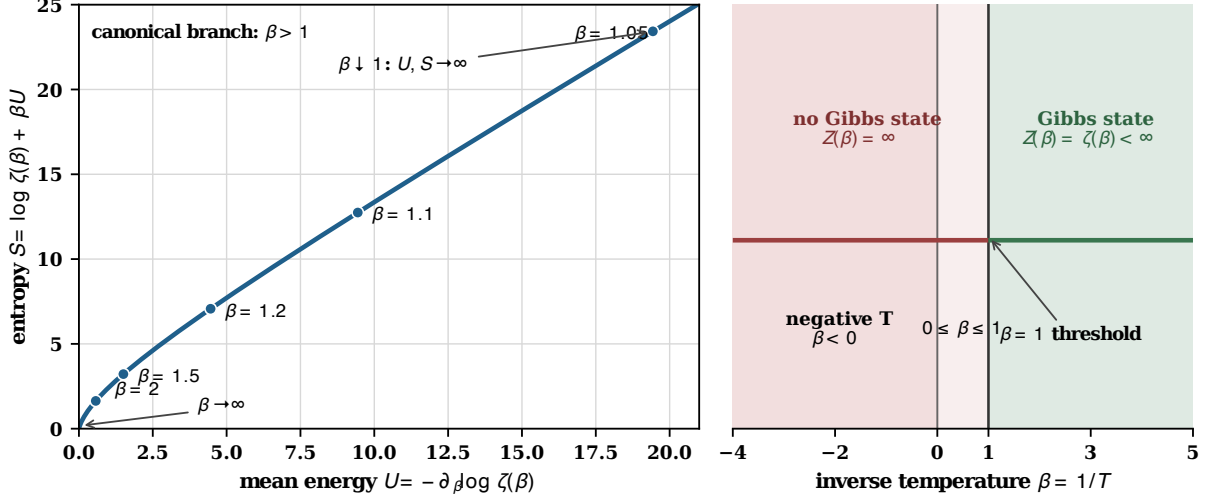


Figure 8: The canonical entropy–energy curve of the zeta Gibbs system and its inverse-temperature domain. Since the spectrum $E_n = \log n$ is unbounded above, $\sum_n e^{-\beta E_n}$ diverges for every $\beta \leq 1$. Thus the model has no normalized negative-temperature branch ($\beta < 0$); analytic continuation of ζ does not supply one.

The obstruction is the divergence of $\sum_n e^{-\beta E_n}$, not the spectrum itself, and completing the zeta function removes it. Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

which is entire and satisfies $\xi(s) = \xi(1-s)$. Riemann’s kernel gives

$$\xi\left(\frac{1}{2} + \beta\right) = \int_{-\infty}^{\infty} \Phi(u)e^{\beta u} du, \quad \Phi(u) = \sum_{n \geq 1} (4\pi^2 n^4 e^{9u/2} - 6\pi n^2 e^{5u/2}) e^{-\pi n^2 e^{2u}}, \quad (22)$$

with $\Phi > 0$, $\Phi(-u) = \Phi(u)$, and doubly exponential decay. So ξ , recentred on the critical line, is the two-sided Laplace transform of a *positive even measure* — a partition function in the strict sense, where ζ alone was not. Three things follow, and Figure 9 shows them. There is no Hagedorn point: $\log Z$ is finite and convex for every real β , so the curve exists on the whole line rather than on a half-line. The excluded negative-temperature branch of Figure 8 is restored, and carries no new information, because Φ is even: $Z(\beta) = Z(-\beta)$, the mean energy is odd and the entropy even. That mirror symmetry *is* the functional equation.

If the exchange-profile formula is applied formally to two individual prime modes, then

$$C_{\text{prof}}(\text{Atom}(p) \rightarrow \text{Atom}(q)) = \inf_{\beta > 0} \frac{-\log(1 - p^{-\beta})}{-\log(1 - q^{-\beta})} = \begin{cases} 1, & p \leq q, \\ 0, & p > q. \end{cases} \quad (23)$$

Indeed, for $p < q$ the ratio is at least one and tends to one as $\beta \downarrow 0$; for $p > q$ it tends to zero as $\beta \rightarrow \infty$. This degenerate answer is informative: over the full temperature range, unequal infinite ladders do not have matching low- and high-temperature asymptotics. A nontrivial comparison needs a temperature window, a work offset, or a different operational model. Equation (23) is therefore a profile calculation, not a new coding theorem for primes.

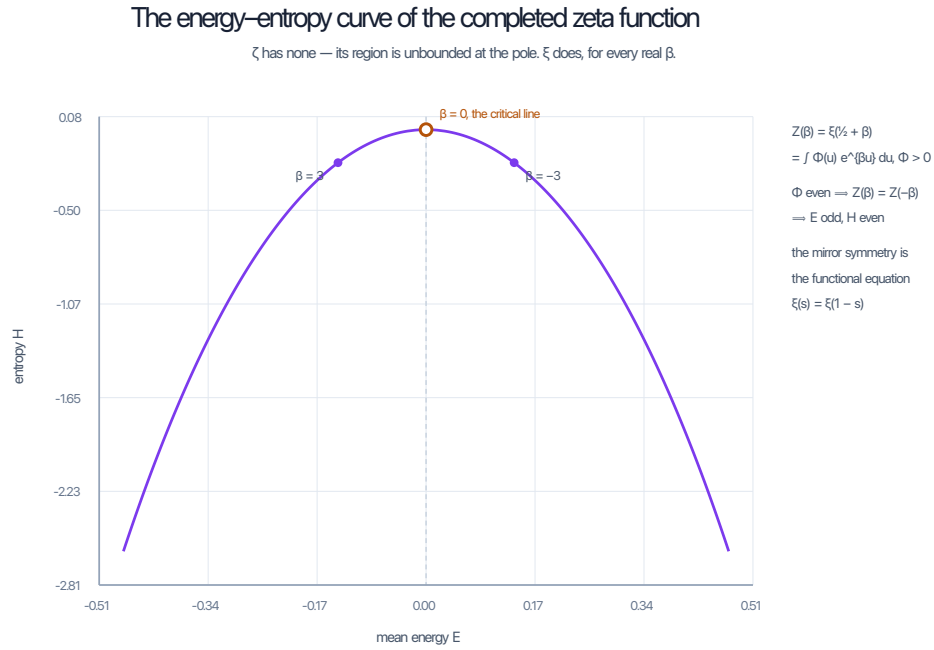


Figure 9: The entropy–energy curve of the completed zeta function, $Z(\beta) = \xi(\frac{1}{2} + \beta)$. Compare Figure 8: there the curve exists only for $\beta > 1$ and diverges at the Hagedorn point, here it exists for every real β and is bounded. The reflection about the entropy axis is the functional equation $\xi(s) = \xi(1-s)$, so negative temperatures are no longer excluded — they are the mirror image of positive ones. Convexity of $\log Z$ was checked numerically throughout the plotted range.

12.1 Complex temperature and zeros

For $\beta > 1$, normalize the Gibbs law

$$\Pr_{\beta}(N = n) = \frac{n^{-\beta}}{\zeta(\beta)}.$$

Then

$$\frac{\zeta(\beta + it)}{\zeta(\beta)} = \mathbb{E}_{\beta}[e^{-it \log N}] \quad (24)$$

is the characteristic function of the random log-energy. A zero in this half-plane would be exact destructive phase cancellation, but the Euler product proves that no such zero exists.

The nontrivial zeros lie in $0 < \Re s < 1$, outside the normalizable Gibbs regime of (21). Analytic continuation assigns values there, but no positive probability measure $n^{-1/2}/\zeta(1/2)$ exists. Consequently

$$\rho = \frac{1}{2} + i\gamma$$

should not presently be called a literal complex energy level. A safer statement is that γ is a logarithmic Mellin frequency: terms such as $x^{\rho} = x^{1/2}e^{i\gamma \log x}$ oscillate in log scale in explicit formulas.

The quantum-chaos analogy reverses the roles. It seeks an unknown Hermitian Hamiltonian whose eigenvalues are the ordinates γ , while the prime powers p^k contribute periodic-orbit lengths $k \log p$. Berry and Keating's $H = xp$ program is a prominent version of this analogy [18]. It does not currently identify a zero with a quantum composition of the independent prime modes in (21).

12.2 What the completion adds

The completed function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

has three conceptually different pieces.

1. $\zeta(s) = \prod_p(1 - p^{-s})^{-1}$ is the product of the finite-prime local factors.
2. $\pi^{-s/2}\Gamma(s/2)$ is the archimedean local factor, obtained as the Mellin transform of a Gaussian.
3. $s(s-1)/2$ cancels the poles at 0 and 1, producing an entire function.

If

$$\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t},$$

then for $\Re s > 1$

$$\int_0^{\infty} (\theta(t) - 1)t^{s/2} \frac{dt}{t} = 2\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Poisson summation gives $\theta(t) = t^{-1/2}\theta(1/t)$, and splitting the integral at $t = 1$ yields analytic continuation and the reflection

$$\xi(s) = \xi(1-s).$$

In Tate's adelic formulation, the finite and archimedean factors are local Mellin transforms and the functional equation is global Fourier duality [16]. For the Gibbs analogy, this is the main lesson of the completion: the global object is not only a product of positive local oscillators; it is completed by a self-dual archimedean mode and a scale reflection.

13 The Weil matrix E and the rate matrix M

For an admissible test profile g with Mellin transform $\widehat{g}$, the Weil quadratic form may be written schematically as

$$Q_W(g) = \sum_{\rho} \widehat{g}(\rho) \overline{\widehat{g}(1 - \bar{\rho})},$$

with the standard symmetric convergence. Polarization gives $B_W(g, h)$, and a finite family gives the Hermitian matrix

$$E_{ij} = B_W(g_i, g_j).$$

Weil’s explicit-formula criterion says that positivity for every admissible family is equivalent to the Riemann hypothesis [17]. Under RH, $1 - \bar{\rho} = \rho$, so E becomes a Gram matrix of the sampled Mellin transforms.

This matrix and the operational exchange matrix have different definitions:

	rate matrix M	Weil matrix E
input	maps and allowed processors	admissible Mellin test profiles
direction	directed	Hermitian
composition law	$M_{ij}M_{jk} \leq M_{ik}$	sesquilinear polarization
positivity notion	no profitable cycle	positive semidefinite quadratic form
arithmetic data	implementation counts	primes, gamma factor, and zeros
known bridge	none in this paper	none in this paper

One exploratory construction multiplies a map profile $\mathcal{Z}_i(s)$ by a log-Gaussian Mellin window and samples the first N known critical-line zeros. With $\sigma = 1/20$, omitting a common positive factor, the real matrix is

$$E_{ij}^{(N)} = 2 \sum_{n=1}^N e^{-\gamma_n^2/400} \operatorname{Re} \left(\overline{\mathcal{Z}_i(\rho_n)} \mathcal{Z}_j(\rho_n) \right).$$

For zeros already sampled on the critical line this is visibly a Gram matrix, so its positive semidefiniteness is automatic. For $N = 100$, the observed ranks are:

family	nodes	rank
quadratic ternary forms over $\mathbb{F}_3$	5	5
quadratic homogeneous $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	7	6
displayed quadratic representatives over $\mathbb{Q}_3$	8	5
displayed quadratic representatives over $\mathbb{Q}_2$	12	5

The rank losses come from exact profile identities or repeated local zeta profiles. They say nothing yet about affine conversion. In particular, positive semidefiniteness of these finite-height Gram matrices is not evidence that no-arbitrage implies Weil positivity, and it is not an independent test of RH.

To connect M and E , one would need at least:

1. a canonical assignment from polynomial-map resources to admissible adelic test profiles;
2. compatibility of that assignment with processor composition and Cartesian powers;

3. a theorem translating operational inequalities into the sesquilinear explicit formula, or conversely.

No such construction is presently known here.

14 What is classical, what is new, and what is open

The main contribution is a new angle and a reproducible synthesis, not a claim to have rediscovered quadratic-form theory. The following ledger states the current novelty boundary.

topic		prior status	contribution and claim level
left–right equivalence		classical in singularity and orbit theory	use the same processor equation simultaneously for equivalence, exact singular reachability, and block exchange
quadratic normal forms		classical Witt, discriminant, radical, and Arf theory	a uniform resource-oriented derivation of the nodes and arrow direction over finite, real, complex, and local fields
finite-field counts	quadratic	consequences of classical orbit–stabilizer calculations	closed formulas and a reproducible atlas of class-sized Hasse diagrams for all odd q , all even $q \geq 4$, and the exceptional $q = 2$ case; the exact compilation appears to be new, but not the ingredients
homogeneous and cubic cases		instances belong to invariant theory and finite computation	candidate new computational datasets: the $\mathbb{F}_3$ five-, seven-, fifty-, twenty-six-, and nineteen-class diagrams, and the 110-class generated preorder over $\mathbb{F}_8$
asymptotic rates		resource theories and asymptotic spectra are classical; the signature formula is proved in the companion paper	attach the solved signature rate to polynomial-map orbit nodes while explicitly separating it from the affine operational rate
cycle return		cycle means and max–plus algorithms are classical	apply them to map-class exchange matrices as round-trip-loss and observer-resolution diagnostics
certified fractional affine rate		Sylvester’s law and the behaviour of rank and inertia under pullback are classical	the exact operational value $C_{\text{aff}}(x^2 - y^2 \rightarrow x^2 + y^2) = 1/2$, the three-copies-for-two certificate in the opposite direction, and the resulting certified one-way pair; elementary ingredients, but we have not seen the statement in operational form
signature rates of curve fibrations		the extreme trace, the second moment, permutation polynomials, and the $USp(2g)$ edge law are classical	no new arithmetic: the rates are monotone functions of statistics already studied, and we say so (Remark 11.6). Offered instead: the closed forms, the family-independent bottleneck $\beta^* = \sqrt{2} - 1$, and two invisibility statements that are about the comparison, not the curves
zeta as partition function		Euler product and Bost–Connes are classical	place the prime-mode Gibbs picture next to map profiles and show exactly where the analogy ceases to be an ordinary Gibbs system
RH and Weil positivity		classical Weil criterion	no new RH claim; the missing M -to- E bridge is formulated as an open problem

Three items are the strongest candidates for genuinely new results after a full literature review:

1. the explicit all- q affine-processor Hasse atlas for degree-at-most-two maps $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q$, with orbit sizes at every node;
2. the exhaustive homogeneous and cubic finite-field datasets and their verification programs;
3. the combined study of the exact degeneration poset, the solved fiber-signature exchange observer, and the max-times cycle losses on the same named classes.

These should be advertised as new computations and organization unless a prior source is found. The no-arbitrage proof itself is elementary and should not be sold as a deep new theorem; its value is that it places a precise constraint on every rate matrix generated by the framework.

15 Open problems

Question 15.1 (Affine coding theorem). Can $C_{\text{aff}}(g \rightarrow f)$ be characterized by a complete family of algebraic or analytic monotones, analogous to the partition functions that solve the finite-signature theory?

Question 15.2 (The definite-to-hyperbolic rate). Is $C_{\text{aff}}(x^2 + y^2 \rightarrow x^2 - y^2) = 2/3$? Corollary 9.6 proves $2/3 \leq C_{\text{aff}} \leq 1$ over $\mathbb{R}$, and searches over $\mathbb{R}$ and $\mathbb{F}_3$ find nothing above $2/3$, but rank and inertia are both spent: an upper bound below one requires a monotone that is not simply subadditive under sums of pullbacks. Is the paired spanning-tree construction of Proposition 9.5 optimal, and does the same value hold over every field on which the two classes are distinct?

Question 15.3 (The seven 2-adic anisotropic classes). Are distinct anisotropic binary forms over $\mathbb{Q}_2$ asymptotically interconvertible at a rate strictly greater than $1/2$? Does any pair have rate one without a finite rate-one conversion?

Question 15.4 (Cycle separation). For which processor categories does a uniform gap $r(\mathcal{C}) \leq 1 - \varepsilon$ over all nontrivial cycles characterize an asymptotic separation of equivalence classes? When does a cycle mean of one imply asymptotic reversibility rather than merely a coarse observer?

Question 15.5 (A cycle among arithmetic families). Do three families of plane-curve fibrations over a fixed $\mathbb{F}_q$ ever satisfy $f_1 \prec f_2 \prec f_3 \prec f_1$ for the signature comparison $a \prec b \iff C_{\text{sig}}(a \rightarrow b) < C_{\text{sig}}(b \rightarrow a)$? If both rates of a pair are attained at an endpoint then $a \prec b \iff \varphi(a) < \varphi(b)$ with $\varphi = \log(\#\text{fibers}) \cdot \log(\max \text{fiber})$, so the comparison is a total preorder there and no cycle is possible; a cycle therefore needs a pair on which an interior tangency overturns φ , and for curve fibrations the interior correction is only $O(1/(q \log q))$. A positive answer would show that the comparison of arithmetic families is not a function of any single real invariant, which is the one statement in Section 11 that would not be a repackaging of a classical statistic (Remark 11.6). A negative answer should be quantitative: how large an interior correction is needed to overturn a typical φ -gap?

Question 15.6 (Functorial zeta profiles). Is there a lattice-independent or adelic profile assignment that is monotone under affine implementation, multiplicative under Cartesian products, and rich enough to bound or determine C_{aff} ?

Question 15.7 (The M -to- E bridge). Can a natural completion of polynomial-map resources produce admissible Weil test functions in a way that turns no-arbitrage or conversion monotonicity into positivity of a Hermitian form? A successful answer must explain why a directed max-times matrix should control a sesquilinear matrix; superficial entrywise transformations are insufficient.

16 Conclusion

The equation $f = h_{\text{out}} \circ g \circ h_{\text{in}}$ is deliberately small. With invertible affine processors it gives classical equivalence classes. With singular affine processors it gives a degeneration preorder and Hasse diagrams. With Cartesian powers it gives exchange rates, directed triangle inequalities, and no-arbitrage cycle bounds. With the fiber signature observer it gives partition functions and Gibbs curves already solved in the companion paper.

Quadratic maps show the benefit of the framework because the answer is recognizable: rank, radical, isotropy, discriminant, and the Arf/trace invariant reappear as resource monotones and branching rules. The same language continues from finite fields to real, complex, and p -adic analytic maps, and from scalar quadratics to homogeneous tensors and cubics.

The number-theoretic continuation is promising but incomplete. The Euler product really is a joint partition function of prime occupation modes for $\Re s > 1$; the completed zeta function really is assembled from finite and archimedean local factors and Fourier–Mellin duality. But the nontrivial zeros are not ordinary Gibbs energy levels, and the operational matrix M is not the Weil matrix E . The present contribution is to make the common language precise enough that a future bridge, if it exists, will have a clear mathematical statement.

A A visual atlas over finite fields

The formulas in Section 4 say that all odd finite fields share one six-node shape, while all even fields of order at least four share one seven-node shape. The diagrams below retain the shape and change the orbit sizes. They are included because the class-size labels make the orbit formulas concrete, and because the exceptional $\mathbb{F}_2$ collapse is easiest to remember visually.

B Two cubic processor choices

The next pair shows why the processor class is part of the theorem. The underlying universe is the same set of 6,561 cubic polynomial functions $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$. Affine input processors retain fourteen classes. The preorder generated by quadratic input processors collapses them to three.

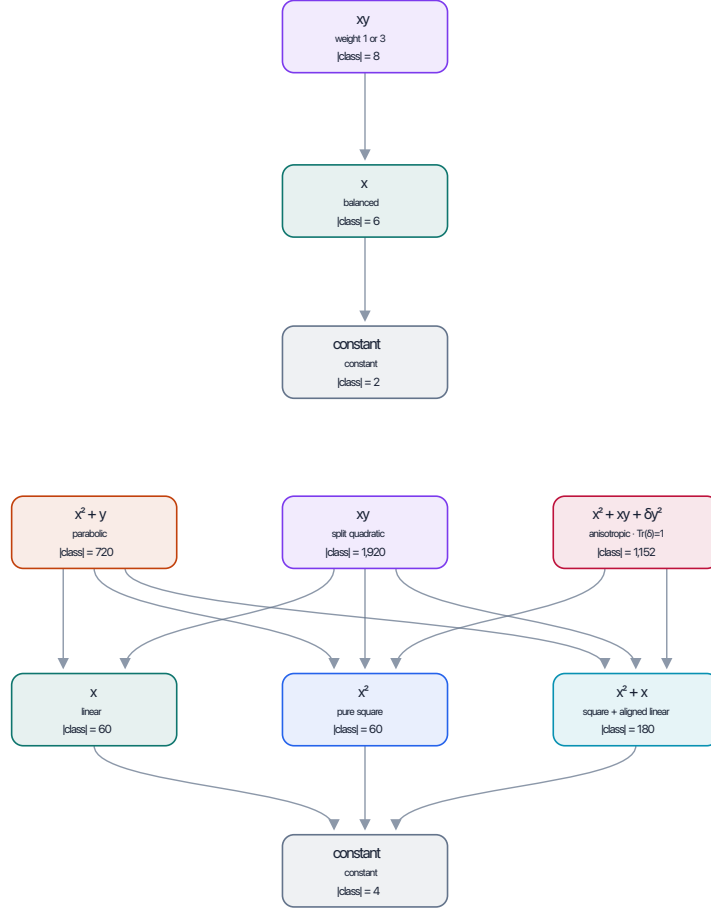


Figure 10: The exceptional three-class, two-cover function poset over $\mathbb{F}_2$, followed by the generic seven-class, eleven-cover poset over $\mathbb{F}_4$. Both use possibly singular affine processors.

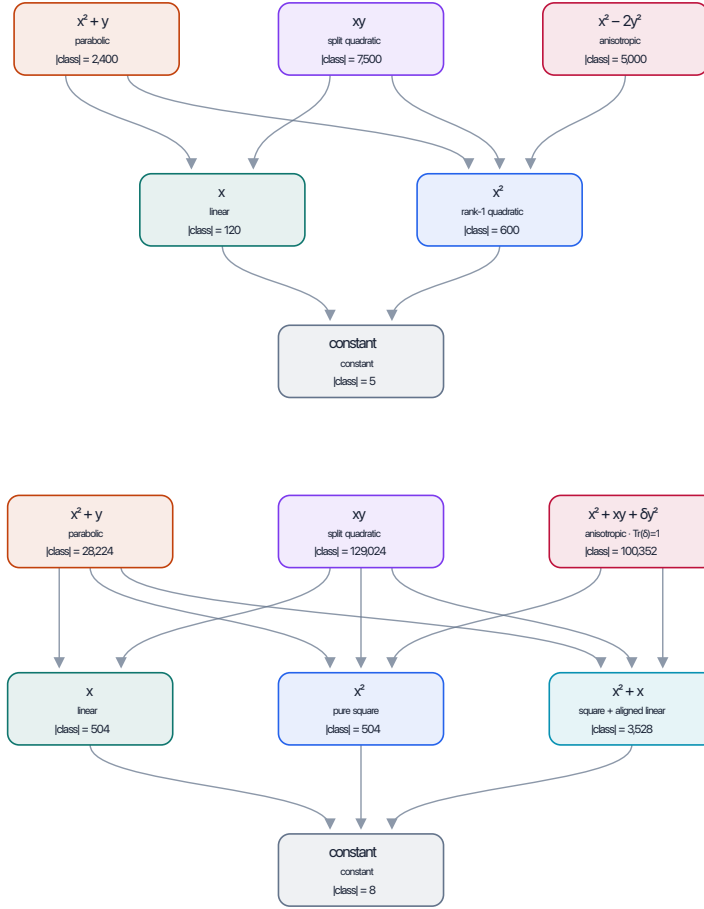


Figure 11: Quadratic maps with singular affine processors over $\mathbb{F}_5$ (six classes, seven covers) and $\mathbb{F}_8$ (seven classes, eleven covers). The field parity changes both the normal forms and the poset shape.

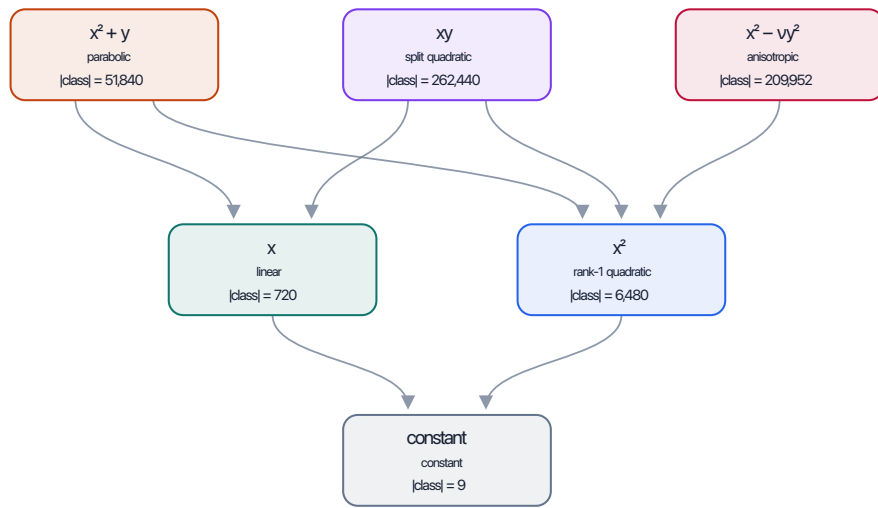


Figure 12: Quadratic maps over $\mathbb{F}_9$ with singular affine processors: six classes and seven covers. The node labels record the field-dependent orbit sizes.

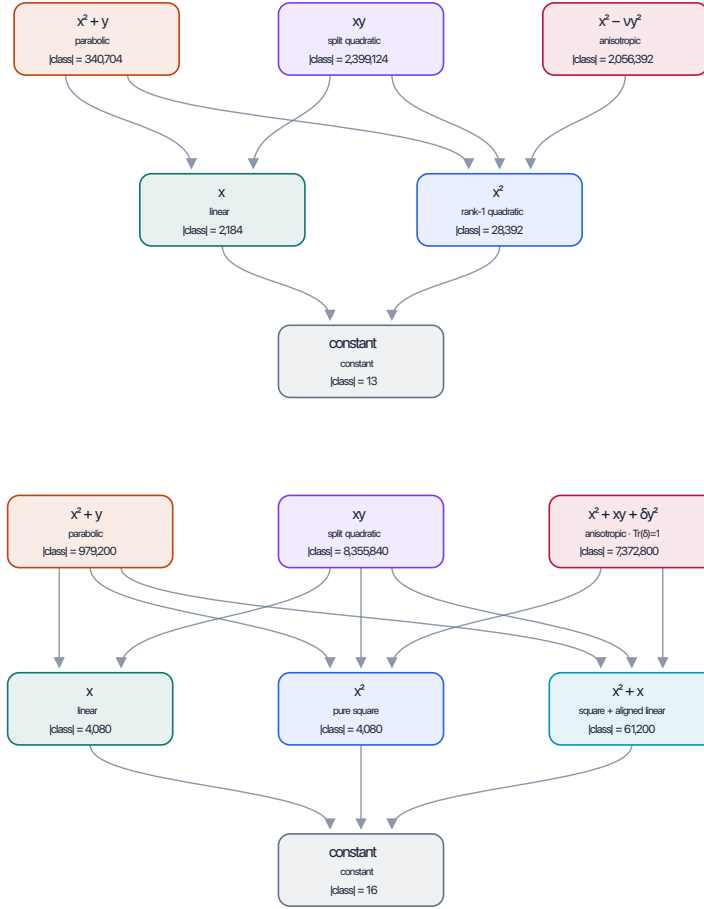


Figure 13: Quadratic maps with singular affine processors over $\mathbb{F}_{13}$ (six classes, seven covers) and $\mathbb{F}_{16}$ (seven classes, eleven covers).

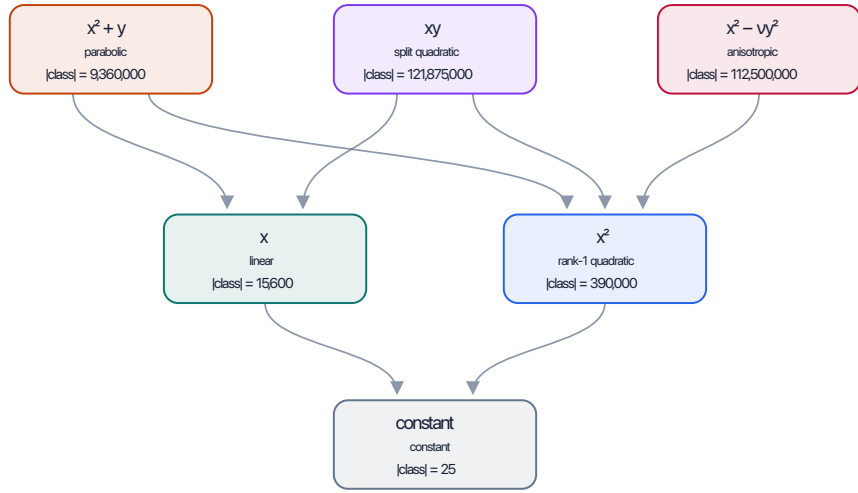


Figure 14: The odd-field diagram over $\mathbb{F}_{25}$, containing $25^6 = 244,140,625$ coefficient vectors in six affine classes and seven covers; singular affine processors define the order.

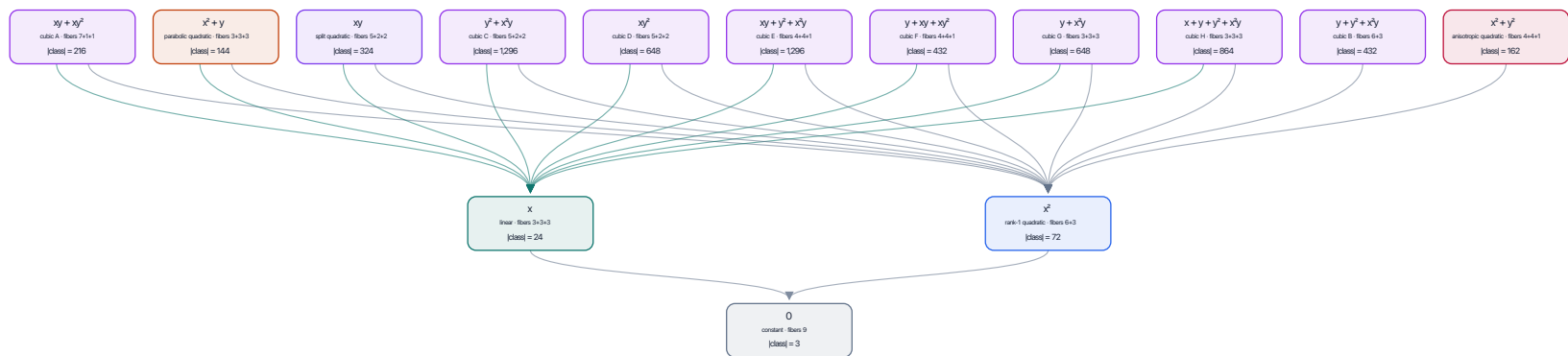


Figure 15: Cubic functions over $\mathbb{F}_3$ with affine input and output processors: fourteen equivalence classes and twenty-two Hasse covers.



Figure 16: The generated preorder with quadratic input and linear output processors: three mutual-reachability classes and two covers. It is not a group-orbit classification by invertible quadratic substitutions.

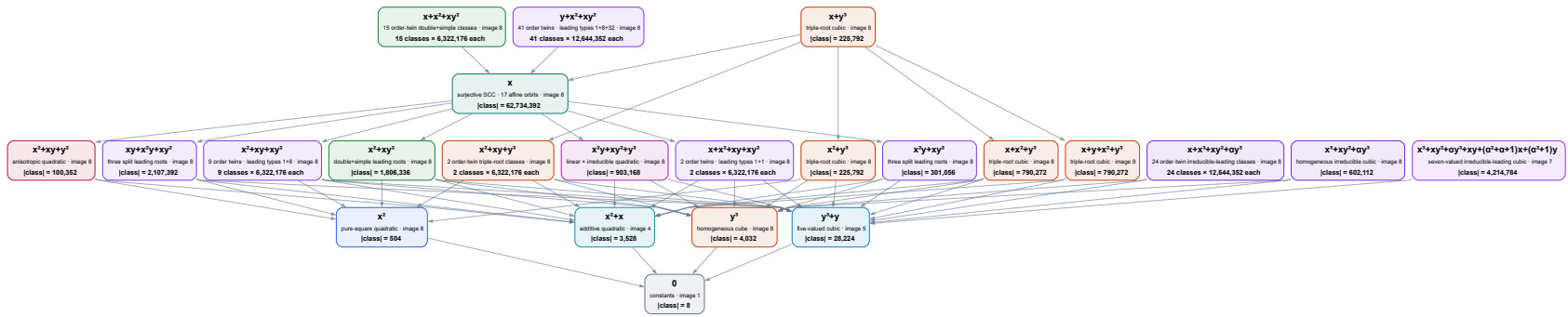


Figure 17: The corresponding quadratic-input calculation over $\mathbb{F}_8$: 110 classes grouped into 23 boxes of order-theoretic twins, with 57 covers between the boxes. Output processors are linear.

Code and data availability

The diagrams, exhaustive classifiers, residue calculations, exchange-rate library, and cycle tables are contained in the accompanying repository. The main regeneration commands are:

- `cli/finite_field_map_poset_cli` for the affine quadratic atlas;
- `cli/homogeneous_tensor_poset_cli` for the homogeneous cases;
- `cli/cubic_map_poset_cli` for the cubic processor comparisons;
- `analysis/polynomial_map_exchange_examples.py` for the exchange matrices and cycle means used here;
- `analysis/definite_hyperbolic_affine_rates.py` for the three-to-two certificate, the exhaustive $\mathbb{F}_3$ searches, and the real feasibility scan of Section 9;
- `analysis/frobenius_bottleneck.py` and `analysis/frobenius_exchange_rates.py` for the closed forms, the bottleneck temperature, and the flatness check of Section 11.

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